RK3572 Android 16.0 SDK Developer Guide

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Preface

Overview

Product Version

Chip Name	Kernel Version
RK3572	kernel-6.12

Intended Audience

This document (this guide) is mainly intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Change Description
V0.0.1	Wu Liangqing	2026-03-02	Released RKR0.1 SDK for RK3572, which is intended for hardware debugging only and not for mass production
V1.0.0	Wu Liangqing	2026-04-16	Released RKR4 SDK for RK3572
V1.1.0	Wu Liangqing	2026-06-10	Updated to RKR5, support RK3572S, changed kernel standalone build method

Document Issue Feedback: wlq@rock-chips.com

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1. RK3572 Android 16.0 SDK Supported Chip

Chip Platform	Support Status	SDK Version
RK3572	Supported	RKR0.1
RK3572	Supported	RKR4
RK3572	Supported	RKR5
RK3572S	Supported	RKR5

2. RK3572 Android 16.0 SDK Source Code Download and Compilation

2.1 Code Download

2.1.1 Download Address

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo
-u https://gerrit.rock-chips.com:8443/Android_16/manifests -b rk3572 -m
Android16.xml
```

For Express permissions, the download address is as follows:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo
-u https://gerrit.rock-chips.com:8443/Android_16/manifests -b rk3572 -m
Android16_Express.xml
```

2.1.2 Server Image Download

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo
-u https://gerrit.rock-chips.com:8443/Android_16/manifests -b rk3572 -m
Android16.xml --mirror
```

For Express permissions, the download address is as follows:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo
-u https://gerrit.rock-chips.com:8443/Android_16/manifests -b rk3572 -m
Android16_Express.xml --mirror
```

Note: repo is a Python script developed by Google to interface with Git, primarily used for downloading and managing Android source repositories. Its download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

To facilitate customers in quickly obtaining the SDK source code, Rockchip technical window typically provides the initial compressed package for the corresponding SDK version. Taking Rockchip_Android16.0_SDK_RELEASE.tar.gz.* as an example, after copying the initial package, the source code can be checked out via the following command:

```
mkdir rk3572_Android16.0_SDK_RELEASE
cat rk3572_Android16.0_SDK_RELEASE.tar.gz* | tar -zx -C
rk3572_Android16.0_SDK_RELEASE
cd rk3572_Android16.0_SDK_RELEASE
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

2.2 Set Up Your Own Repo Code Server

2.2.1 Environment

Install `openssh-server` for remote login, `git` for project management, and `keychain` for public-private key management.

```
sudo apt-get install openssh-server git keychain
```

2.2.2 Gitolite Setup

2.2.2.1 Server-Side Operations

(Taking the server address as an example)

1. Create Git account:

```
sudo adduser --system --shell /bin/bash --group git
sudo passwd git
```

2. Log in as the 'git' account
3. Ensure that "~/.ssh/authorized_keys" is empty or does not exist.
4. Copy the server administrator's public key to "~/YourName.pub"
5. Download gitolite source code

```
git clone https://github.com/sitaramc/gitolite.git
```

6. Create the bin directory in the git user directory

```
mkdir -p ~/bin
```

7. Execute the following command to install gitolite and the installation methods vary by version. Refer to the documentation in the source code:

```
gitolite/install -to ~/bin
```

8. Set Up Administrator

```
~/bin/gitolite setup -pk YourName.pub
```

2.2.2.2 Client-side Operations

1. Clone the gitolite admin repository on the server:

```
git clone server_address/gitolite-admin.git
```

2. Add user public key to the gitolite directory.

```
cp username.pub keydir/username.pub
```

3. Add admin user

```
vi conf/gitolite.conf
@admin = admin1 admin2 admin3
repo gitolite-admin
RW+     =   @admin
```

2.2.3 repo Mirror Setup

2.2.3.1 Server-side Operations

1. Log in to the server using a Git account
2. Download the repo tool in the root directory

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3. Create the RK_Android16_mirror directory

```
mkdir RK_Android16_mirror
```

4. Enter the RK_Android16_mirror directory

```
cd RK_Android16_mirror
```

5. Download RK Android16 SDK image

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo
-u https://gerrit.rock-chips.com:8443/android_a16/manifests -b rk3572 -m
Android16.xml --mirror
```

6. Create Repository Group Permissions

```
.repo/repo/repo list -n > android_a16.conf
sed -i 's/^/@android_t = RK_Android16_mirror\/&/g' android_a16.conf
```

2.2.3.2 Client-side Operations

1. Copy the server-side android_a16.conf to the client-side `gitolite-admin/conf/`
2. Add Group Permissions

```
vi conf/android_a16.conf
@usergroup = user1 user2 user3
repo @android_a16
R   = @usergroup
RW+ = @admin
```

```
vi conf/gitolite.conf
include "android_a16.conf"
```

3. Create your own manifests repository

```
vi conf/android_a16.conf
@android_a16 = Android_T/manifests_xxx
```

2.2.4 Client-side Operations (Local)

1. Download the manifests_xxx repository on the client.

Download the manifests_xxx.git repository on other client machines

```
git clone server_address/android_a16/manifests_xxx.git
```

2. Download the original manifests repository on the client

```
git clone server_address/android_a16/manifests.git
```

3. Submit the manifest.xml file to the newly created manifest_xxx repository.

Copy files from the original manifests directory to manifests_xxx

```
cd manifests_xxx
cp -rf manifests/*.xml manifests_xxx/
```

View copied file

```
git status

 ├── Android16_Express.xml
├── Android16.xml
├── default_remote.xml
├── default.xml
├── include
│   ├── partner_gms_express.xml
│   ├── partner_modules_express.xml
│   ├── rk3572_ebook_repository.xml
│   ├── rk3572_repository.xml
│   ├── rk_checkout_repository.xml
│   ├── rk_modules_repository.xml
│   ├── rk_system_core.xml
│   └── ssi_core.xml
├── remote.xml
├── remove_16.xml
├── remove_unused_backup.xml
└── remove_unused.xml
```

Local Commit

```
git add -A
git commit -m "init xxx"
```

Push to remote branch

```
git push origin master:master
```

4. Create Your Own Code Download Link

Download the repo tool under the root directory

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

After following the above steps, your code download link is as follows:

```
mkdir Android16
cd Android16
~/repo/repo init -u server_address/android_a16/manifests_xxx.git -m
Android16.xml
```

Specifically:

`server_address` is your server address.

Following the above steps, your own repository server setup is complete, and the repository URL can be shared with colleagues for collaborative work.

2.3 Code Management

After setting up the code server as described above, most code repositories use RK's default branch. For repositories requiring code modifications, refer to the following steps.

2.3.1 Switch to Your Own Branch

1. Navigate to the code repository requiring modification, using the kernel directory as an example

```
cd kernel-6.12
```

2. Switch to a local branch

```
git checkout remotes/m/master -b xxx_branch
```

3. Push xxx_branch branch to the remote server

```
git push rk29 xxx_branch:xxx_branch
```

Here, `rk29` is a remote that can be auto-completed directly via the Tab key.

4. Enter the .repo/manifests directory to modify the branch specified in the manifest

Navigate to the .repo/manifests directory and use `grep kernel` to locate the manifest corresponding to the kernel repository.

```
cd .repo/manifests
```

```
--- a/include/rk_modules_repository.xml
+++ b/include/rk_modules_repository.xml
@@ -10,7 +10,7 @@
   <project path="hardware/rockchip/libgraphicpolicy"
name="rk/hardware/rk29/libgraphicpolicy" remote="rk"
revision="refs/tags/android-16.0-mid-rkr1" />
   <project path="hardware/rockchip/libhwjpeg" name="rk/hardware/rk29/libhwjpeg"
remote="rk" revision="refs/tags/android-16.0-mid-rkr5"/>
   <project path="u-boot" name="rk/u-boot" remote="rk"
revision="refs/tags/android-16.0-mid-rkr5"/>

-  <project path="kernel" name="rk/kernel" remote="rk29"
revision="refs/tags/android-16.0-mid-rkr5"/>

+  <project path="kernel" name="rk/kernel" remote="rk29" revision="xxx_branch"/>

   <project path="bootable/recovery/rkupdate"
name="platform/bootable/recovery/rk_update" remote="rk"
revision="refs/tags/android-16.0-mid-rkr5"/>
   <project path="bootable/recovery/rkutility"
name="platform/bootable/recovery/rk_utility" remote="rk"
revision="refs/tags/android-16.0-mid-rkr5"/>
```

5. Submit the modified manifest to the remote branch

```
git add include/rk_modules_repository.xml
git commit -m "change kernel branch on xxx_branch"
git push origin default:master
```

After submitting the manifests repository, colleagues can synchronize the kernel code of your own branch.

2.3.2 Code Modification Commit

After switching branches as per the above steps, you can commit your changes to your branch and directly push to the xxx_branch.

2.3.3 Synchronize RK's Code

1. Synchronizing RK code requires a sync operation on the server side.

```
cd RK_Android16_mirror
```

```
.repo/repo/repo sync -c
```

2. Merging RK's modifications to manifests on the client side

- Download RK original manifests repository

```
git clone server_address/username/manifests.git
```

Use a diff tool to compare manifests (RK's original) and manifests_xxx (your own), and merge the differences modified by RK into your repository (primarily modifying tags, adding or removing repositories, etc.).

- After verification, push the changes to manifests_xxx.

This step also verifies which repositories were modified, and the merging of these repositories will be performed in the next step.

3. Directories with their own branches must manually merge RK's changes into their own branch.

Take kernel as an example:

- View the current remote branch

```
user@host:~/home1/test2/kernel-6.12$ git branch -av
* android-11.0-mid-rkr7   0bde59fad73a ARM: configs: rockchip_defconfig enable
ION_CMA_HEAP
  xxx_branch              0bde59fad73a ARM: configs: rockchip_defconfig enable
ION_CMA_HEAP
  remotes/m/master        -> rk29/xxx_branch
  remotes/rk29/xxx_branch 0bde59fad73a ARM: configs: rockchip_defconfig enable
ION_CMA_HEAP
```

It can be seen that the current reference points to: `remotes/m/master -> rk29/xxx_branch`

- Create a local branch (based on your remote branch)

```
git checkout remotes/m/xxx_branch -b local_xxx_branch
```

- Verify the current latest TAG released by RK

```
user@host:~/home1/test2/kernel$ git tag | grep rkr
android-10.0-mid-rkr1
android-10.0-mid-rkr10
```

```
android-10.0-mid-rkr11
android-10.0-mid-rkr13
android-10.0-mid-rkr2
android-10.0-mid-rkr3
android-10.0-mid-rkr4
android-10.0-mid-rkr5
android-10.0-mid-rkr5
android-10.0-mid-rkr7
android-10.0-mid-rkr8
android-10.0-mid-rkr9
android-11.0-ebook-rkr1
android-11.0-ebook-rkr2
android-11.0-ebook-rkr3
android-11.0-ebook-rkr4
android-11.0-ebook-rkr5
android-11.0-ebook-rkr6
android-11.0-mid-rkr1
android-11.0-mid-rkr2
android-11.0-mid-rkr3
android-11.0-mid-rkr4
android-11.0-mid-rkr4.1
android-11.0-mid-rkr5
android-11.0-mid-rkr5
android-11.0-mid-rkr7
android-11.0-mid-rkr7-prev
android-11.0-mid-rkr8
android-16.0-mid-rkr0.1
```

The latest tag for Android 16 is `android-16.0-mid-rkr0.1`

- Merge `android-16.0-mid-rkr0.1` into local branch

```
git merge android-16.0-mid-rkr0.1
```

Check for conflicts. Resolve any conflicts first. If no conflicts are found, proceed to the next step.

- push the merged code to the remote branch

```
git push rk29 local_xxx_branch:xxx_branch
```

- All other split directories should be merged and submitted using this method.

2.4 Kernel Code Path Overview

Android 16 SDK uses kernel version **kernel-6.12**.

The kernel configuration for RK3572 adopts a layered structure:

Config file	Location	Description
`rockchip_defconfig`	`kernel-6.12/arch/arm64/configs/rockchip_defconfig`	Rockchip chip base defconfig
`android-16.config`	`kernel-6.12/kernel/configs/android-16.config`	Android generic config
`rk3572.config`	`kernel-6.12/arch/arm64/configs/rk3572.config`	RK3572 chip-level config

Config combination order during compilation: `rockchip_defconfig android-16.config <chip-config>`

Chip	Chip config
RK3572	`rk3572.config`
RK3572S	`rk3572.config` (shared with RK3572)
RK3576	`rk3576.config`

Android 16 supports the 6.12 kernel, with the kernel source code located in the kernel-6.12 directory of the project.

2.5 Code Compilation

2.5.1 Setting Environment Variables

Use the envsetup.sh script to initialize the environment:

```
source build/envsetup.sh
```

2.5.2 Select Target lunch

Before compiling Android, it is necessary to determine a build target (target). The build target reflects the platform you are compiling. Determine your build target by using the lunch command followed by a string representing the target. The following provides a detailed procedure and example to assist you in completing this process.

```
lunch rk3572_a16-bp4a-userdebug
```

The string format for the target is as follows:

```
lunch product_name-release-build_variant
```

The components of this string are as follows:

- product_name is the name of the product you want to build, such as rk3572_a16. You can name the specific product_name according to your device format.

- Set release to bp4a, where bp4a represents the configuration for the current stable version, usable for production. Configuration details are available in build/release/aconfig/bp4a. As Android versions update, the configuration for stable versions evolves. Pay attention to the changes in build/release/aconfig and select the latest configuration for mass production. trunk_staging represents the mainline release configuration, usable for debugging new Android features. Configuration details are available in build/release/aconfig/trunk_staging.
- The build_variant component of the string can be one of the three values in the following table:

Build Type	**Usage**
user	Restricted permissions. Suitable for production environment
userdebug	Similar to "user", but with root access and debugging capabilities, which is the preferred build type for debugging
eng	Development configuration with additional debugging tools

Other Lunch Item Descriptions

Launch item	**Compatible chips**	**Additional notes**
rk3572_a16-bp4a-user	RK3572	Product for Android 16, compatible with RK3572 development board hardware, default corresponding dts file is rk3572-evb1-v10.dts, compiled as user build for production use. Android system supports 64-bit only
rk3572_a16-bp4a-userdebug	RK3572	Product for Android 16, hardware adapted for RK3572 development board, default corresponding dts file is rk3572-evb1-v10.dts, compiled as userdebug build, used for development and debugging. Android system supports 64-bit only
rk3572_ebook-bp4a-user	RK3572	eBook product for Android 16, hardware-compatible with RK3572 development board, default corresponding dts file is rk3572-ebook-a3-v10.dts, compiled as user version for production use. Android system supports 64-bit
rk3572_ebook-bp4a-userdebug	RK3572	eBook product for Android 16, hardware-compatible with RK3572 development board, default corresponding dts file is rk3572-ebook-a3-v10.dts, compiled as debugging version for development and debugging. Android system supports 64-bit
rk3572s_a16-bp4a-user	RK3572S	Product for Android 16, compatible with RK3572S chip, default corresponding dts file is rk3572s-evb1-v10.dts, compiled as user build for production use
rk3572s_a16-bp4a-userdebug	RK3572S	Product for Android 16, compatible with RK3572S chip, default corresponding dts file is rk3572s-evb1-v10.dts, compiled as userdebug build for development and debugging

2.5.3 One-Click Compile Command

```
./build.sh -UKAup
( WHERE: -U = build uboot
       -C = build kernel with Clang
       -K = build kernel
       -A = build android
       -p = will build packaging in IMAGE
       -o = build OTA package
       -u = build update.img
       -v = build android with 'user' or 'userdebug'
       -d = build kernel dts name
       -V = build version
       -J = build jobs
 )

============================================================
Note that environment variables must be set and the target platform for
compilation must be selected before using the one-click build command, for
example:
source build/envsetup.sh
lunch rk3572_a16-bp4a-userdebug
============================================================
```

2.6 Platform-Specific Compilation Commands Summary

Soc	Type	Reference Model	Android	One-Click Build	Kernel Build	U-Boot Build
RK3572	EVB1 Development Board	rk3572-evb1-v10	build/envsetup.sh;lunch rk3572_a16-bp4a-userdebug	./build.sh -AUCKu	./build.sh -K	./build.sh -U
RK3572	E-Book	rk3572-ebook-a3-v10	build/envsetup.sh;lunch rk3572_ebook-bp4a-userdebug	./build.sh -AUCKu	./build.sh -K	./build.sh -U
RK3572S	EVB1 Development Board	rk3572s-evb1-v10	build/envsetup.sh;lunch rk3572s_a16-bp4a-userdebug	./build.sh -AUCKu	./build.sh -K	./build.sh -U

2.7 Generic Kernel Image (GKI)

Products passing GMS and EDLA certification for Android 16 mandate the use of GKI for the kernel. RK3572 SDK does not enable GKI by default. To enable the GKI feature, make the following modifications (using the RK3572 platform as example):

```
user@sys2206:~/b0_A14_bringup/device/rockchip/rk3572$ git diff
diff --git a/rk3572_a16/BoardConfig.mk b/rk3572_a16/BoardConfig.mk
index eb52fb6..f59df62 100755
--- a/rk3572_a16/BoardConfig.mk
+++ b/rk3572_a16/BoardConfig.mk
@@ -14,7 +14,7 @@
 # limitations under the License.
 #
 BUILD_WITH_GO_OPT := false
-BOARD_BUILD_GKI := false
+BOARD_BUILD_GKI := true

 BOARD_GRAVITY_SENSOR_SUPPORT := true
```

Setting BOARD_BUILD_GKI := true automatically enables the A/B feature.

Refer to the documentation RKDocs/android/《Rockchip_Developer_Guide_Android14_GKI_CN》for details on GKI kernel compilation, ko updates, and related procedures.

2.7.1 Other Compilation Notes

2.7.1.1 Android16.0 cannot directly flash kernel.img and resource.img

The following compilation applies only to non-GKI. For GKI, refer to the document RKDocs/android/《Rockchip_Developer_Guide_Android14_GKI_CN》

The kernel.img and resource.img for Android 16.0 are included in boot.img, requiring the use of the build.sh -AK command to compile the kernel. After compilation, flash the boot.img in the rockdev directory. This process recompiles Android, resulting in a longer compilation time. It is recommended to compile the kernel separately as follows.

2.7.1.2 Separate Compilation of Kernel to Generate boot.img

> **Note:** kernel-6.12 requires rust for compilation, different from previous kernel versions. It is recommended to use the `mkboot.sh` script below. This method works for both non-GKI and GKI builds.

Principle of Compilation: Replace the existing `boot.img` with the newly generated `kernel.img` and `resource.img` in the kernel directory.

Using the `RK3572` reference hardware as an example, replace the corresponding boot.img and dts during compilation:

Here, `BOOT_IMG=../rockdev/Image-rk3572_a16/boot.img` specifies the path to the old boot.img, which should be adjusted according to the actual path.

It is recommended to save the following script as `mkboot.sh` in the `kernel-6.12` directory and then run it for compilation.

```
#!/bin/bash
# mkboot.sh - Compile kernel separately to generate boot.img
# Usage: ./mkboot.sh <dts-name> <output-boot.img-path> [chip-config]
# Example: ./mkboot.sh rk3572-evb1-v10 boot_sample.img rk3572.config
```

```
export PATH=$PWD/../prebuilts/clang/host/linux-x86/clang-r574158/bin:$PATH
export PATH=$PWD/../prebuilts/rust/linux-x86/1.88.0/bin:$PATH
CLANG_PATH=$PWD/../prebuilts/clang/host/linux-x86/clang-r574158
RUST_PATH=$PWD/../prebuilts/rust/linux-x86/1.88.0

ADDON_ARGS="CROSS_COMPILE=aarch64-linux-gnu- LLVM=1 LLVM_IAS=1
KBUILD_GENDWARFKSYMS_STABLE=1 ARCH=arm64"
KERNEL_BUILD_TOOL="LD_LIBRARY_PATH=$PWD/../prebuilts/kernel-build-tools/linux-
x86/lib64 PATH=$PWD/../prebuilts/kernel-build-tools/linux-
x86/bin:$PWD/$CLANG_PATH/bin:$PWD/$RUST_PATH/bin:$PATH HOSTCFLAGS=-
I$PWD/../prebuilts/kernel-build-tools/linux-x86/include HOSTLDFLAGS=-
L$PWD/../prebuilts/kernel-build-tools/linux-x86/lib64 LIBCLANG_PATH=\$(dirname
\$(which clang))/../lib BINDGEN=$PWD/../prebuilts/clang-tools/linux-
x86/bin/bindgen"

make $KERNEL_BUILD_TOOL $ADDON_ARGS rockchip_defconfig android-16.config \$3 &&
make $KERNEL_BUILD_TOOL $ADDON_ARGS BOOT_IMG=\$2 \$1.img -j32
```

Compilation command examples:

RK3572 (EVB1):

```
cd kernel-6.12
./mkboot.sh rk3572-evb1-v10 ../rockdev/Image-rk3572_a16/boot.img rk3572.config
```

RK3572S (EVB1):

```
cd kernel-6.12
./mkboot.sh rk3572s-evb1-v10 ../rockdev/Image-rk3572s_a16/boot.img rk3572.config
```

> Note: Different chips require different config files, generally named .config, e.g., `rk3572.config`, `rk3576.config`, `rk3588.config`.

Compile and directly flash the generated boot.img in the kernel-6.12 directory to the device's boot partition. **Please load the partition table (parameter.txt) first** to avoid incorrect flashing location.

2.8 Firmware Flashing

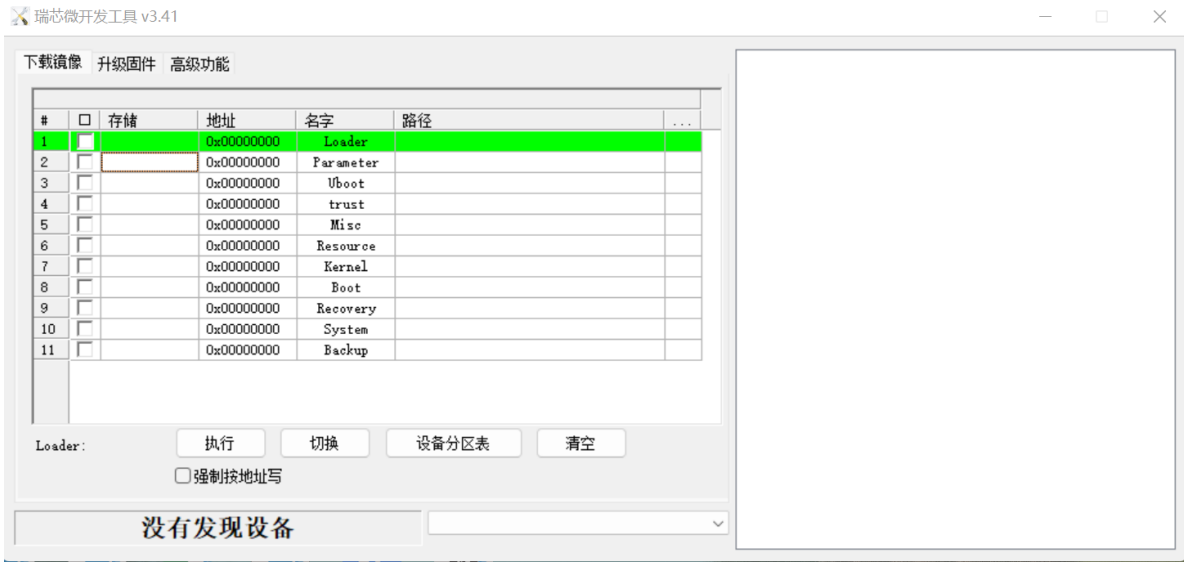
2.8.1 Firmware Flashing Tool

RK3572 Android 16's USB driver DriverAssitant must be updated to version V5.14, and the update can be performed by referring to the Tools section below.

Windows Flashing Tool: (The tool is constantly updated. Please synchronize updates promptly.)

RK3572 platform Windows firmware flashing tool must use version V3.41 or later.

```
RKTools\windows\AndroidTool/RKDevTool_v3.41_for_window
```



Linux Flashing Tool:

RK3572 platform Linux firmware flashing tool must use version V2.55 or later.

```
RKTools/linux/Linux_Upgrade_Tool/upgrade_tool_v2.55_for_linux
```

Refer to the details in the following Tool Description section.

2.8.2 Firmware Notes

Firmware	Description
boot.img	contains ramdisk, kernel, dtb
boot-debug.img	The difference from boot.img is that user firmware can flash this boot.img to gain root access.
dtbo.img	Device Tree Overlays: Refer to the dtbo section below
config.cfg	Configuration file for the flashing tool, directly importable to display the flashing options
MiniLoaderAll.bin	Contains primary loader
misc.img	Contains recovery-wipe boot flag information, and recovery is triggered after flashing.
parameter.txt	contains partition information
pcba_small_misc.img	Contains pcba boot identification information. After flashing, enters minimal PCBA mode
pcba_whole_misc.img	Contains PCBA boot identification information. After flashing, enters full PCBA mode
recovery.img	contains recovery-ramdisk, kernel, and dtb
super.img	contains the contents of the odm, product, vendor, system, and system_ext partitions
trust.img	Contains BL31, BL32. RK3566/RK3568 do not generate this firmware, and it does not need to be flashed.
uboot.img	contains U-Boot firmware
vbmeta.img	Contains AVB verification information for AVB verification
update.img	Contains the img files required for direct flashing of the entire firmware package via tools.

2.9 fastboot flashing dynamic partitions

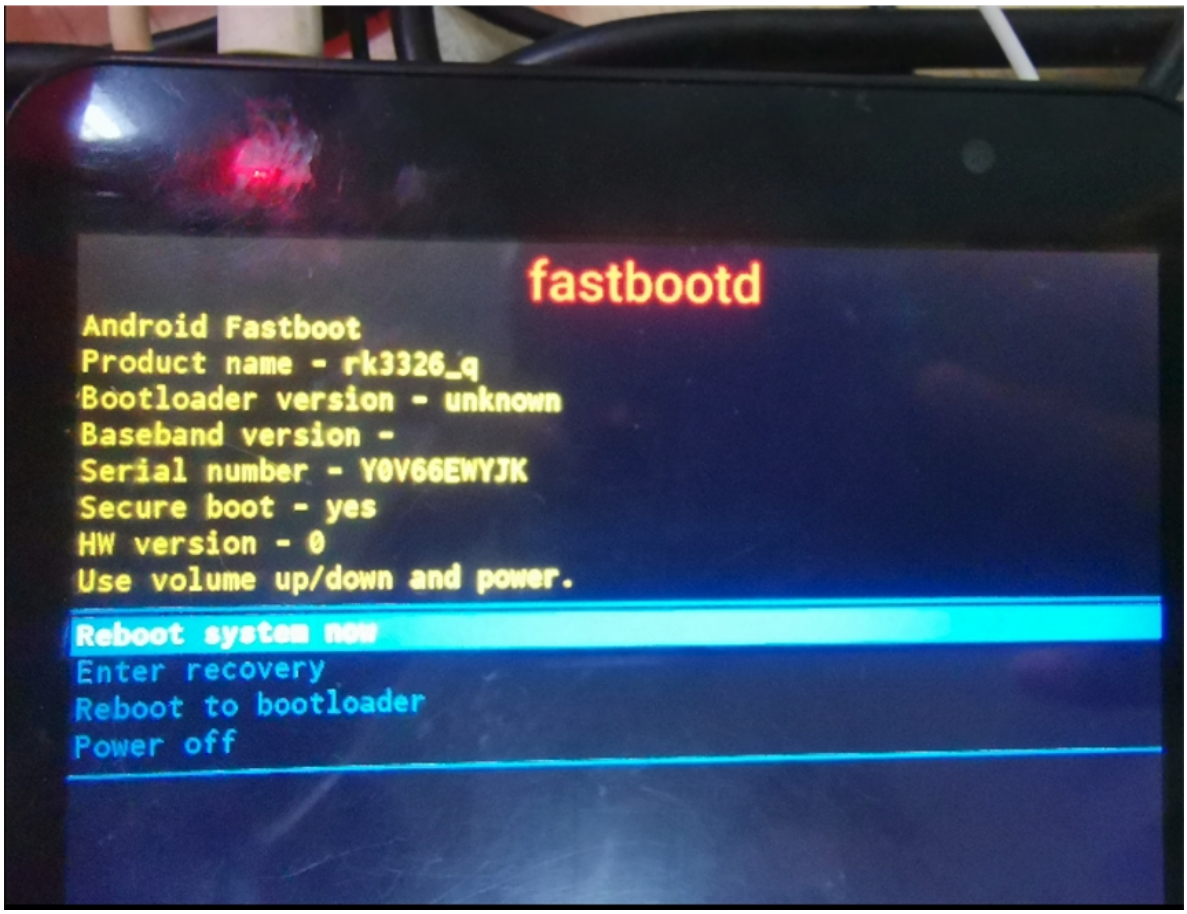
R introduces support for dynamic partitions, having removed the system/vendor/odm/product/system_ext partitions. Please flash super.img. You can use `fastbootd` to flash system/vendor/odm individually (the corresponding img files can be found in the out directory), requiring both adb and fastboot to be at the latest version. The SDK provides prebuilt toolchains:

```
RKTools/linux/Linux_adb_fastboot (Linux_x86 version)
RKTools/windows/adb_fastboot (Windows_x86 version)
```

- Use the command to flash dynamic partitions:

```
adb reboot fastboot
fastboot flash vendor vendor.img
fastboot flash system system.img
fastboot flash odm odm.img
```

Note: After entering fastbootd mode, the screen displays relevant device information, as shown in the figure:



Note: To use fastboot for non-dynamic partitions, please enter bootloader:

```
adb reboot bootloader
```

The method to flash GSI:

After confirming the device is unlocked, enter fastbootd mode and flash only system.img from GSI and misc.img from the firmware. After flashing, the device will enter recovery mode for factory reset. The complete flashing procedure is provided below.

1. Reboot to bootloader, locked → device unlocked:

```
adb reboot bootloader
fastboot oem at-unlock-vboot ## For OEMs who have flashed the AVB public key,
refer to the corresponding documentation for unlocking.
```

2. Factory reset, reboot to fastbootd:

```
fastboot flash misc misc.img
fastboot reboot fastboot  ## Enters fastbootd mode
```

3. Begin flashing GSI

```
fastboot delete-logical-partition product  ## Optional: For devices with
constrained partition space, execute this command to delete the product
partition before flashing GSI
fastboot flash system system.img
fastboot reboot  ## After successful flashing, reboot
```

- Note: GSI can also be flashed using DSU (Dynamic System Updates). The Rockchip platform now defaults to supporting DSU. Due to the significant memory consumption, it is not recommended for devices with 1GB DDR or lower. Refer to the Android documentation for details on DSU usage and implementation.

https://source.android.com/devices/tech/ota/dynamic-system-updates

- Note 1: During VTS testing, the compiled boot-debug.img must be simultaneously flashed to the boot partition.
- Note 2: CTS-ON-GSI testing does not require flashing boot-debug.img.
- Note 3: During testing, use the GSI image officially released by Google with a -signed suffix.

3. Utilizing DTBO Feature

Android 10.0 and above supports the Device Tree Overlays feature, where the development process involves flashing dtbo.img for multi-product compatibility, etc.

Modification Method:

1. Locate (or specify) the template file:

```
get_build_var PRODUCT_DTBO_TEMPLATE
```

For example:

```
PRODUCT_DTBO_TEMPLATE := $(LOCAL_PATH)/dt-overlay.in
(device/rockchip/rk3572/rk3572_a16/dt-overlay.in)
```

2. Add or modify the required nodes:

For example:

```
/dts-v1/;
/plugin/;

&chosen {
    bootargs_ext = "androidboot.boot_devices=${_boot_device}";
};

&firmware_android {
    vbmeta {
        status = "disabled";
    };
    fstab {
        status = "disabled";
    };
```

```
};

&reboot_mode {
    mode-bootloader = <0x5242C309>;
    mode-charge = <0x5242C30B>;
    mode-fastboot = <0x5242C303>;
    mode-loader = <0x5242C301>;
    mode-normal = <0x5242C300>;
    mode-recovery = <0x5242C303>;
};
```

Note: When using **dtbo**, ensure that an **alias** exists in the **dts**, otherwise, the **overlay** will fail.

4. Modify fstab file

1. Locate (or specify) the template file:

```
get_build_var PRODUCT_FSTAB_TEMPLATE
```

For example:

```
PRODUCT_FSTAB_TEMPLATE := device/rockchip/common/scripts/fstab_tools/fstab.in
```

2. Changes: Add partition mount, modify swap_zram parameter, and modify data partition format, etc.

5. Modify parameter.txt

Android 14 adds a tool to generate parameter.txt, supporting generation based on configuration parameters. If no template file is configured, the tool will look for and add a modified parameter.txt file.

1. Locate (or specify) the template file:

```
get_build_var PRODUCT_PARAMETER_TEMPLATE
```

For example:

```
PRODUCT_PARAMETER_TEMPLATE :=
device/rockchip/common/scripts/parameter_tools/parameter.in
```

2. Modify configuration partition size (e.g.):

```
BOARD_SUPER_PARTITION_SIZE := 2688548864
BOARD_DTBOIMG_PARTITION_SIZE := xxxx
BOARD_BOOTIMAGE_PARTITION_SIZE := xxxxx
BOARD_CACHEIMAGE_PARTITION_SIZE := xxxx
```

3. Do not use parameter generation tool:

Add a parameter.txt file to your device directory:

For example: device/rockchip/rk3572/rk3572_a16/parameter.txt

4. Only use the tool to generate parameter.txt (for example):

```
parameter_tools --input
device/rockchip/common/scripts/parameter_tools/parameter.in --firmware-version
16.0 --machine-model rk3572 --manufacturer rockchip --machine rk3572_a16 --
partition-list
uboot_a:4096K,trust_a:4M,misc:4M,dtbo_a:4M,vbmeta_a:4M,boot_a:33554432,backup:30
0M,security:4M,cache:300M,metadata:4096,frp:512K,super:2G --output
parameter_new.txt
```

Note: For major version OTA upgrades, please directly use the previous version's parameter.txt

5. Add a new partition

Newly create baseparameter partition as an example:

- Define BOARD_WITH_SPECIAL_PARTITIONS in the product's BoardConfig.mk

like: BOARD_WITH_SPECIAL_PARTITIONS := baseparameter:1M,logo:16M

```
device/rockchip/rk3572/rk3572_a16/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -494,4 +494,11 @@ ifeq ($(strip $(BOARD_TWRP_ENABLE)), true)
+BOARD_WITH_SPECIAL_PARTITIONS := baseparameter:1M
```

- Add BOARD_WITH_SPECIAL_PARTITIONS to RebuildParameter.mk

```
device/rockchip/common/build/rockchip/RebuildParameter.mk
+ifneq ($(strip $(BOARD_WITH_SPECIAL_PARTITIONS)), )
+partition_list := $(partition_list),$(BOARD_WITH_SPECIAL_PARTITIONS)
+endif
```

6. Common Android Configuration

6.1 Create Product lunch

- Navigate to device/rockchip/rk3572 directory.
- Create a new directory named rk3572_new_a16 in the device/rockchip/rk3572 directory.
- Create based on the existing rk3572_a16 product directory in the device/rockchip/rk3572 directory. First, copy the rk3572_a16 directory, then rename it to rk3572_new_a16.

You can use the following command to copy the directory:

```
cp -r rk3572_a16 rk3572_new_a16
```

- After copying the directory, navigate to the rk3572_new_a16 directory.

Replace the string rk3572_a16 with rk3572_new_a16 in all files within the rk3572_new_a16 directory. You may use tools such as sed or manually edit the files.

For example, you can use the following sed command:

```
sed -i 's/rk3572_a16/rk3572_new_a16/g' *
```

This command will replace rk3572_a16 with rk3572_new_a16 in all files within the rk3572_new_a16 directory.

- After modifying the files, your new product directory rk3572_new_a16 is ready for further customization and development.

7. Kernel dts Instruction

7.1 Create Product dts

New product dts can be created by selecting the corresponding dts as a reference based on the configuration in the following table.

7.2 DTS Structure (RK3572 & RK3572S)

Kernel DTS source code is located in `kernel-6.12/arch/arm64/boot/dts/rockchip/` and adopts a **chip-level .dtsi → board-level .dtsi → product-level .dts** layered structure.

RK3572 DTS Hierarchy (using EVB1 as example):

```
rk3572.dtsi               ← Chip-level: defines CPU, GIC, CRU, GRF, VOP, HDMI,
display subsystem, etc.
  ├── rk3572-evb.dtsi     ← Board-level: EVB common peripherals (buttons, ADC,
I2C, etc.)
  │     └── rk3572-evb1-v10.dtsi    ← Board variant: EVB1 specific config
(audio, camera, display)
  │           └── rk3572-evb1-v10.dts  → Product DTS (entry file)
  │
  ├── rk3572-rk806.dtsi   ← PMIC RK806 configuration
  └── rk3572-pinctrl.dtsi ← Pin multiplexing configuration
```

`rk3572-evb1-v10.dts` include chain:

```
#include "rk3572-evb1-v10.dtsi"    // Board-level config
#include "rk3572-android.dtsi"     // Android system config
#include "rk3572-evb1-cam-csi0.dtsi"  // Camera config
```

RK3572S DTS Hierarchy:

```
rk3572.dtsi               ← RK3572 chip-level (shared with RK3572)
  └── rk3572s.dtsi        ← RK3572S diff layer (delete/modify unsupported
modules)
        └── rk3572s-evb.dtsi    ← RK3572S EVB board-level
              └── rk3572s-evb1-v10.dtsi  ← RK3572S EVB1 board variant
                    └── rk3572s-evb1-v10.dts  → Product DTS (entry file)
```

Main differences between RK3572S and RK3572 (in `rk3572s.dtsi`):

- Removed Ethernet (gmac0/gmac1), PCIe1, SATA0/SATA1, NPU
- Removed CAN0-CAN3, DSMC, EBC, VOPL
- Removed vp1 (only vp0 retained)
- Display subsystem ports only retain `<&vop_out>` (vopl removed)
- PD_NPU removed pm_perf/pm_qos configuration
- PD_VOP pm_qos only retains `<&qos_vop_m0>` (m1 removed)

Soc	PMIC	Development Board Type	Model	DTS
RK3572	RK806	Development Board	RK3572 EVB1	rk3572-evb1-v10.dts
RK3572	RK806	Development Board	RK3572 EVB2	rk3572-evb2-v10.dts
RK3572	RK809+RK805	tablet prototype	RK3572 TABLET	rk3572-tablet-v10.dts
RK3572	RK806	E-Book	RK3572 EBook	rk3572-ebook-a3-v10.dts
RK3572S	RK806	Development Board	RK3572S EVB1	rk3572s-evb1-v10.dts

8. Patch Release

Important patches are occasionally released on the redmine system, with links as follows:

```
https://redmine.rock-chips.com/projects/rockchip_patch/issues
```

Real-time email notifications for patch releases can be obtained via subscription. The subscription method is as follows:

Step 1: Log in to the Redmine system

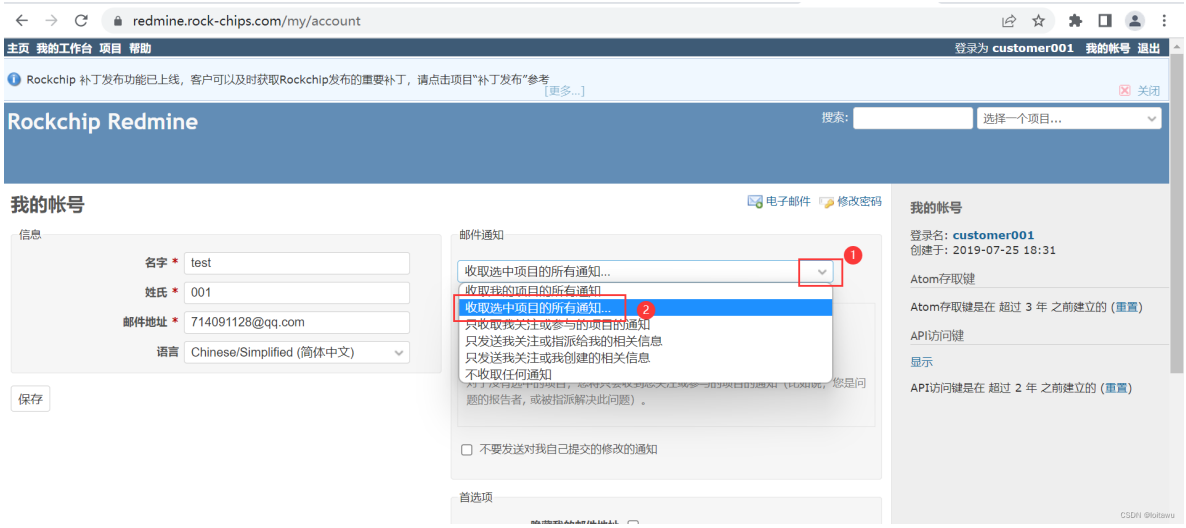
Use the Redmine account registered with Rockchip to log in to the Redmine system.

Step 2: Access My Account



Step 3: Select email notification type

As shown in the figure, select **Collect all notifications for selected items** from the **Email Notifications** dropdown.



Step 4: Select Project

Select the **Patch Release** item as shown in the figure and click **Save**



The above steps complete the subscription to **patch release**.

Upon successful subscription, email notifications are received via the email address registered on Redmine when Rockchip releases patches.

9. Document Overview

9.1 Peripherals Support List

The support list for DDR/EMMC/NAND FLASH/WIFI/3G/CAMERA is updated in real-time on redmine, with links as follows:

```
https://redmine.rockchip.com.cn/projects/fae/documents
```

9.2 Camera IQ Tool Documentation

```
external/camera_engine_rkaiq/rkisp2x_tuner/doc/
├── Rockchip_Color_Optimization_Guide_ISP2x_CN_v2.0.0.pdf
├── Rockchip_IQ_Tools_Guide_ISP2x_CN_v2.0.0.pdf
└── Rockchip_Tuning_Guide_ISP21_CN_v2.0.0.pdf
```

9.3 rknn-toolkit2 Development SDK and Documentation

```
hardware/rockchip/rknn-toolkit2/doc/
```

9.4 RKDocs Documentation Guide

```
RKDocs/
├── android
│   ├── Android Dual Display Developer Guide.zip
│   ├── audio
│   │   ├── Rockchip_Developer_Guide_Android_3A_Module_CN.pdf
│   │   ├── Rockchip_Developer_Guide_Android_Audio_AIDL_HAL_CN.pdf
│   │   ├── Rockchip_Developer_Guide_Android_EQ_DRC_CN.pdf
│   │   ├── Rockchip_Developer_Guide_Android_Multi_Audio_CN.pdf
│   │   └── Rockchip_Developer_Guide_Automotive_Audio_CN.pdf
│   ├── bt
│   │   └── Rockchip_Introduction_Android9.0_BT_Configuration_CN.pdf
│   ├── patches
│   │   └── root
│   │       ├── Rockchip_Introduction_Android_Root_CN.pdf
│   │       ├── Rockchip_Introduction_Android_Root_EN.pdf
│   │       └── RootChecker.apk
│   ├── project.config
│   ├── Rockchip_Android_GMS_Media_FAQ_CN.pdf
│   ├── Rockchip_Android_Remote_key_Provisioning_Guide.pdf
│   ├──
Rockchip_Developer_Guide_AB_System_OTA_from_Android14_to_Android16_CN.pdf
│   ├── Rockchip_Developer_Guide_Android_Optimization_CN.pdf
│   ├── Rockchip_Developer_Guide_Android14_GKI_CN.pdf
│   ├── Rockchip_Developer_Guide_Android14_GKI_EN.pdf
```

```
│   ├── Rockchip_Developer_Guide_Android16_SSI_SDK_CN.pdf
│   ├── Rockchip_Developer_Guide_Android16_SSI_SDK_EN.pdf
│   ├── Rockchip_Developer_Guide_Android_AB_System_Upgrading_CN.pdf
│   ├── Rockchip_Developer_Guide_Android_Automotive_CN.pdf
│   ├── Rockchip_Developer_Guide_Android_GRF_OTA_CN.pdf
│   ├── Rockchip_Developer_Guide_Android_Recovery_CN.pdf
│   ├── Rockchip_Developer_Guide_Android_SELinux(Sepolicy)_CN.pdf
│   ├── Rockchip_Developer_Guide_PCBA_Test_Tool_V1.3_CN&EN.pdf
│   ├── Rockchip_Firmware_Upgrade_Failed_Analyze_Method_CN.pdf
│   ├── Rockchip_Introduction_Android_Application_Preinstallation_CN&EN.pdf
│   ├── Rockchip_Introduction_Android_Boot_Video_CN.pdf
│   ├── Rockchip_Introduction_Android_BOX_Display_Framework_Configuration_CN.pdf
│   ├── Rockchip_Introduction_Android_Factory_Reset_Protection_CN&EN.pdf
│   ├── Rockchip_Introduction_Android_Log_System.pdf
│   ├── Rockchip_Introduction_Android_Performance_Mode_CN&EN.pdf
│   ├── Rockchip_Introduction_Android_Power_On_Off_Animation_and_Tone_Customization_CN&EN.pdf
│   ├── Rockchip_Introduction_Android_Samba_CN.pdf
│   ├── Rockchip_Introduction_Android_Widevine_Project_Start_Preparation_CN.pdf
│   ├── Rockchip_Introduction_Box_Media_Application_CN&EN.pdf
│   ├── Rockchip-Parameter-File-Format-Version1.4-CN.pdf
│   ├── Rockchip_User_Guide_Android_Device_ID_Attestation_Flash_CN.pdf
│   ├── Rockchip_User_Guide_Android_GMS_Configuration_CN.pdf
│   ├── Rockchip_User_Guide_Android_GMS_Configuration_EN.pdf
│   ├── Rockchip_User_Guide_Box_FactoryTestTool_V3.0_CN.pdf
│   ├── Rockchip_User_Guide_Dr.G_CN&EN.pdf
│   ├── Rockchip_User_Guide_Magisk_Installation_EN.pdf
│   ├── video
│   │   ├── Rockchip_Android_Introduction_Libhwjpeg_Interface_CN.pdf
│   │   ├── Rockchip_Android_Introduction_Libhwjpeg_Interface_EN.pdf
│   │   ├── Rockchip_Android_Multimedia_FAQ_CN.pdf
│   │   ├── Rockchip_Android_Multimedia_FAQ_EN.pdf
│   │   ├── Rockchip_Android_Widevine_L1_Guide.pdf
│   │   ├── Rockchip_Developer_Guide_Android_Boot_Video_CN.pdf
│   │   ├── Rockchip_Developer_Guide_Android_Boot_Video_EN.pdf
│   │   ├── Rockchip_Introduction_Android_MediaCodec_Vendor_Parameters_CN.pdf
│   │   ├── Rockchip_Introduction_Android_MediaCodec_Vendor_Parameters_EN.pdf
│   │   └── Rockchip_Introduction_Android_Smart_Encoder_CN.pdf
│   └── wifi
│       ├── Rockchip_Introduction_REALTEK_WIFI_Driver_Porting_CN&EN.pdf
│       └── Rockchip_Introduction_WIFI_Configuration_CN&EN.pdf
└── common
    ├── Audio
    │   ├── Rockchip_Developer_Guide_Audio_Call_3A_Algorithm_Integration_and_Parameter_Debugging_CN.pdf
    │   ├── Rockchip_Developer_Guide_Audio_CN.pdf
    │   ├── Rockchip_Developer_Guide_RK817_RK809_Codec_CN.pdf
    │   └── Rockchip_User_Guide_Android14_Audio_Function_CN.pdf
    ├── camera
    │   ├── common
    │   │   └── Camera_External_FAQ_v1.0 .pdf
    │   ├── HAL1
    │   │   ├── README_CN.txt
    │   │   ├── README_EN.txt
    │   │   ├── RK312x_Camera_User_Manual_v1.4(3288&3368).pdf
    │   │   ├── RK_ISP10_Camera_User_Manual_v2.3.pdf
```

```
    │   │   ├── RKISPV1_Camera_Module_AVL_v1.7.pdf
    │   │   ├── Rockchip_Camera_AVL_v2.0_Package_20180515.7z
    │   │   ├──
Rockchip_Introduction_RKISPV1_Camera_Driver_Debugging_Method_CN.pdf
    │   │   ├── Rockchip_Introduction_RKISPV1_Camera_FAQ_CN.pdf
    │   │   ├── Rockchip SOFIA 3G-
R_PMB8018(x3_C3230RK)_Camera_Module_AVL_v1.6_20160226.pdf
    │   │   └── Rockchip_Trouble_Shooting_Android_CameraHAL1_CN_EN.pdf
    │   ├── HAL3
    │   │   ├── camera_engine_rkisp_user_manual_v2.2.pdf
    │   │   ├── camera_hal3_user_manual_v2.3.pdf
    │   │   ├── README_CN.txt
    │   │   ├── RKCIF_Driver_User_Manual_v1.0.pdf
    │   │   ├── RKISP1_IQ_Parameters_User_Guide_v1.2.pdf
    │   │   ├── RKISP_Driver_User_Manual_v1.3.pdf
    │   │   ├── Rockchip_Camera_Module_OTP_Calibration_Guide
    │   │   │   └── ISP30
    │   │   │       ├── CN
    │   │   │       │   └── Rockchip Camera Module OTP Calibration Guide_CN.pdf
    │   │   │       └── EN
    │   │   │           └── Rockchip Camera Module OTP Calibration Guide_EN.pdf
    │   │   ├── Rockchip_Color_Optimization_Guide_ISP
    │   │   │   ├── ISP21
    │   │   │   │   └── CN
    │   │   │   │       └──
Rockchip_Color_Optimization_Guide_ISP21_CN_v2.0.1.pdf
    │   │   │   ├── ISP30
    │   │   │   │   ├── CN
    │   │   │   │   │   └──
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10. Tool Usage

10.1 StressTest

Use the Stresstest tool on the device to perform stress testing on various functions of the device under test, ensuring the stability of the entire system operation. The SDK can launch the StressTest application by opening the calculator app and entering the secret code `"83991906="` for stress testing on various functions.

Stresstest testing tool test content mainly includes:

10.1.1 Module-Related

- Camera stress testing: includes camera on/off, photo capture, and camera switching.
- Bluetooth stress testing: includes Bluetooth on/off.
- Wi-Fi stress testing: includes Wi-Fi on/off, (ping tests and iperf tests to be added).

10.1.2 Non-Module-Related

- Airplane Mode On/Off Stress Test
- Sleep/Resume Stress Test
- Video Stress Test
- Reboot Stress Test
- Factory Reset Stress Test
- ARM Frequency Scaling Test

- GPU Frequency Scaling Test
- DDR Frequency Scaling Test

10.2 PCBA Test Tools

PCBA test tools are used to quickly assess product functionality during mass production, enhancing production efficiency. Current test items include LCD, Wi-Fi, Bluetooth, DDR/EMMC storage, SD card (SD card), USB Host, key (KEY), and Codec.

These test items include automatic test items and manual test items, with wireless network, DDR/EMMC, and Ethernet as automatic test items, and buttons, SD card, USB HOST, Codec as manual test items.

For specific PCBA functional configuration and usage instructions, please refer to:

```
RKDocs/android/Rockchip_Developer_Guide_PCBA_Test_Tool_CN&EN.pdf_V1.1_20171222.pdf
```

10.3 DeviceTest

DeviceTest is used for factory-level final assembly testing, primarily verifying the functionality of peripheral components after complete assembly. The SDK accesses DeviceTest by opening the calculator application and entering the secret code "000.=", as shown below:



On the production line, the corresponding peripheral components can be tested according to this interface. During testing, click "TEST CHECKED" to test each item individually. Click 'Pass' for successful tests and 'Failed' for failed tests. The final results will be displayed on the interface, as shown in the following figure.

Items marked in red are failed, while others are passed. The factory can perform appropriate repairs based on the test results. Additionally, if the customer requires further customization on the tool, please contact the FAE window to apply for the corresponding source code.

10.4 USB Driver

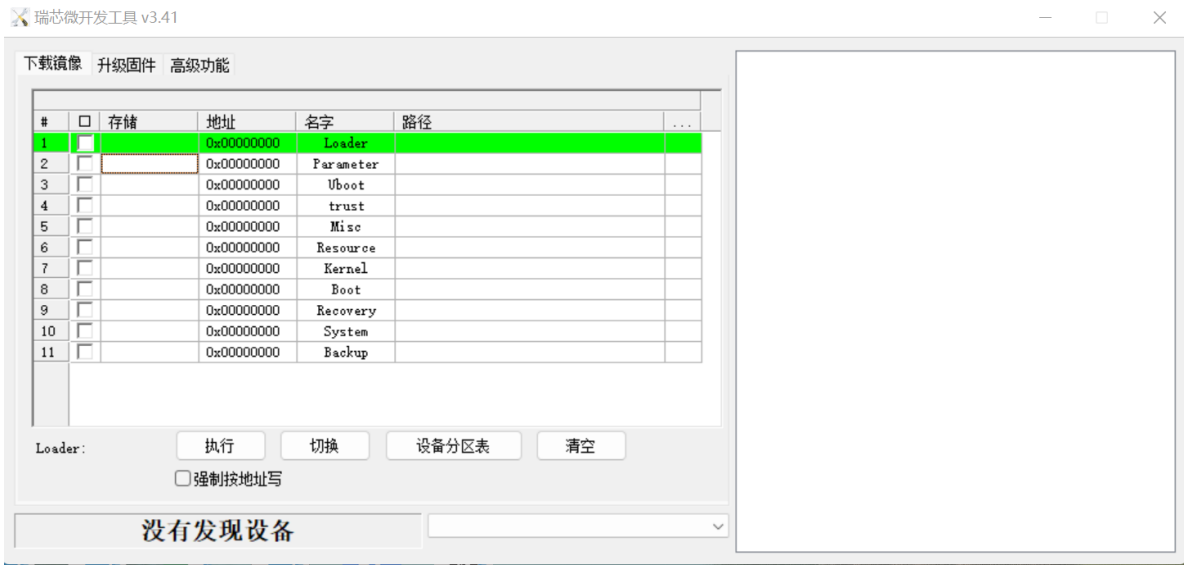
Rockchip USB driver package includes ADB and firmware flashing driver.

```
RKTools\windows\DriverAssitant_v5.14.zip
```

10.5 Development Flashing Tool

10.5.1 Windows Version

RKTools/windows/AndroidTool/AndroidTool_Release_v3.41.zip, the tool version is continuously updated. Please keep it updated promptly.



10.5.2 Linux Version

RKTools/linux/Linux_Upgrade_Tool/Linux_Upgrade_Tool_v2.55.zip

```
Linux_Upgrade_Tool_v2.26$ sudo ./upgrade_tool -h
Program Data in ~/.config/upgrade_tool

----------------------Tool Usage ----------------------
Help:              H
Quit:              Q
Version:           V
Clear Screen:      CS
------------------Upgrade Command ------------------
ChooseDevice:            CD
ListDevice:                  LD
SwitchDevice:            SD
UpgradeFirmware:         UF <Firmware> [-noreset]
```

```
UpgradeLoader:          UL <Loader> [-noreset]
DownloadImage:          DI <-p|-b|-k|-s|-r|-m|-u|-t|-re image>
DownloadBoot:           DB <Loader>
EraseFlash:             EF <Loader|firmware> [DirectLBA]
PartitionList:          PL
WriteSN:                SN <serial number>
ReadSN:             RSN
----------------Professional Command -----------------
TestDevice:             TD
ResetDevice:            RD [subcode]
ResetPipe:              RP [pipe]
ReadCapability:         RCB
ReadFlashID:            RID
ReadFlashInfo:          RFI
ReadChipInfo:           RCI
ReadSector:             RS  <BeginSec> <SectorLen> [-decode] [File]
WriteSector:            WS  <BeginSec> <File>
ReadLBA:                RL  <BeginSec> <SectorLen> [File]
WriteLBA:               WL  <BeginSec> <File>
EraseLBA:               EL  <BeginSec> <EraseCount>
EraseBlock:             EB <CS> <BeginBlock> <BlokcLen> [--Force]
-------------------------------------------------------
```

10.6 SD Upgrade Boot Creation Tool

Used for SD card upgrade, SD card boot, and SD card PCBA testing.

```
RKTools\windows\SDDiskTool_v1.74.zip
```

10.7 Number Writing Tool

RKTools\windows\RKDevInfoWriteTool-1.3.0.7z

Extract RKDevInfoWriteTool-1.3.0.7z and install

Run the software as administrator

Refer to the tool documentation for details:

```
RKDocs/common/RKTools manuals\Rockchip_User_Guide_RKDevInfoWriteTool_CN.pdf
```

10.8 DDR Soldering Test Tool

To test DDR hardware connections and diagnose soldering defects such as cold joints.

```
RKTools\windows\Rockchip_Platform_DDR_Test_Tool_V1.38_Release_Annoucement_CN.7z
RKTools\windows\Rockchip_Platform_DDR_Test_Tool_V1.38_Release_Annoucement_EN.7z
```

10.9 efuse Flashing Tool

efuse flashing for RK3288W/RK3368/RK3399 platforms.

```
RKTools\windows\efuse_v1.37.rar
```

10.10 efuse/otp Signing Tool

Firmware eFuse/OTP Signature

```
RKTools\windows\SecureBootTool_v1.94.zip
```

10.11 Factory Production Firmware Flashing Tool

Used for factory batch firmware flashing.

```
RKTools\windows\FactoryTool-1.72.9.7z
```

10.12 Data Preloading Tool for userdata Partition

Tool for generating userdata partition preloaded data packages.

```
RKTools\windows\OemTool_v1.3.rar
```

10.13 Camera IQ Tool

Used for debugging ISP image effects.

```
external/camera_engine_rkaiq/rkisp2x_tuner
```

11. System Debugging

11.1 ADB Tool

11.2 Overview

ADB (Android Debug Bridge) is a tool included in the Android SDK that enables the management and operation of Android emulators or physical Android devices. Key features include:

- Run the device's shell (command-line)
- Manage port mappings for simulators or devices
- Upload/download files between computers and devices

- Install the local APK on an emulator or Android device

ADB is a "client-server" program, where the client primarily refers to a PC, and the server runs on the physical Android device or a virtual machine. Depending on the method used to connect the PC to the Android device, ADB can be categorized into two types:

- ADB over Network: Host connects to STB device via wired/wireless network (same local network)
- USB ADB: Host connects to STB device via USB cable

11.2.1 USB ADB Usage Guide

USB ADB has the following limitations:

- supports only USB OTG port
- does not support multiple clients operating simultaneously (e.g., cmd windows, Eclipse, etc.)
- Only supports connecting a single device to the host and does not support connecting multiple devices

Connection steps are as follows:

1. The Android device is running the Android system, with Settings > Developer Options > Connected to Computer enabled and USB debugging turned on.
2. The PC host connects only via a USB cable to the device's USB OTG port, then uses the following command to connect to the Android device.

```
adb shell
```

3. Verify the connection by running the `adb devices` command. A successful connection is indicated by the display of the device's serial number.

11.3 Common ADB Commands Instruction

1. View device status

View the Android devices or emulators connected to the PC:

```
adb devices
```

The returned result is the serial number of the Android device connected to the development machine, or the IP address and port number (Port), status.

2. Install APK

Install the specified APK file on the device:

```
adb install <APK file path>
```

For example:

```
adb install “F:\WishTV\WishTV.apk”
```

Reinstall the application:

```
adb install -r "F:\WishTV\WishTV.apk"
```

3. Uninstall APK

Completely uninstall:

```
adb uninstall <package>
```

For example:

```
adb uninstall com.wishtv
```

4. Use rm to remove APK files:

```
adb shell rm <filepath>
```

For example:

```
adb shell rm "system/app/WishTV.apk"
```

Example: Remove the "WishTV.apk" file from the "system/app" directory.

5. Access the shell of the device and emulator

Enter the shell environment of the device or emulator:

```
adb shell
```

6. Upload files from computer to device

The push command can upload any file or directory from the local PC to the device. The local path generally refers to the local PC, while the remote path typically refers to the single-board device connected via adb.

adb push

Examples are as follows:

```
adb push "F:\WishTV\WishTV.apk" "system/app"
```

Example: Upload the local "WishTV.apk" file to the Android system's "system/app" directory.

7. Download files from device to computer

The pull command can download files or folders from the device to the local PC.

```
adb pull <remote path> <local path>
```

For example:

```
adb pull system/app/Contacts.apk F:\
```

Example: Download files or folders from the Android system "system/app" directory to the local "F:\" directory.

8. View bug report

To view all system-generated error message reports, run the `adb bugreport` command, which will display the information of Android system dumpsys, dumpstate, and logcat.

9. View Device System Information

The specific command to view device system information under adb shell.

```
adb shell getprop
```

11.4 Logcat Tool

The Android logging system provides functionality for recording and viewing system debugging information. Logs are recorded from various software components and certain system buffers, which can be viewed and utilized via Logcat. Logcat is the most frequently used feature for debugging programs. This functionality primarily displays program runtime behavior by printing logs. Given the extensive volume of logs to be printed, filtering and other operations are required.

11.4.1 Logcat Command Usage

Use the logcat command to view the contents of the system log buffer:

Basic Format:

```
[adb] logcat [<option>] [<filter-spec>]
```

For example:

```
adb shell
logcat
```

11.5 Common Log Filtering Methods

Methods for Controlling Log Output:

- Control log output priority

For example:

```
adb shell
logcat *:W
```

Example: Display log messages at warning level or higher.

- Control log tags and output priority

For example:

```
adb shell
logcat ActivityManager:I MyApp:D *:S
```

Example explanation: All log entries are supported except those with tag "ActivityManager" and priority "Info" or higher, and those with tag "MyApp" and priority "Debug" or higher.

- Only output logs with specific tags

For example:

```
adb shell
logcat WishTV:* *:S
```

or

```
adb shell
logcat -s WishTV
```

Example: Output only logs tagged as WishTV.

- Only output logs with specified priority and tag

For example:

```
adb shell
logcat WishTV:I *:S
```

Example: Output only logs with priority I and label WishTV.

11.6 Procrank Tool

Procrank is a built-in debugging tool on Android, running in the device-side shell environment to output memory snapshots, enabling efficient monitoring of process memory usage.

Includes the following memory information:

- VSS: Virtual Set Size virtual memory usage size (including memory occupied by shared libraries)
- RSS: Resident Set Size actual physical memory usage size (including memory occupied by shared libraries)
- PSS: Proportional Set Size actual physical memory size used (proportionally allocated for shared libraries)
- USS: Unique Set Size, the size of physical memory exclusively occupied by the process (excluding memory occupied by shared libraries)

Note:

- USS size refers to the memory exclusively used by the current process, which is fully reclaimed upon process termination.
- VSS/RSS includes memory used by shared libraries, which is not relevant for examining a single process's memory state.
- PSS represents the proportional share of shared memory occupied by a single process following the proportional division of the shared memory pool.

11.6.1 Using procrank

Root access must be granted to the terminal before executing procrank.

su

Command format:

```
procrank [ -W ] [ -v | -r | -p | -u | -h ]
```

Common Commands Overview:

-v: sort by VSS

-r: Sort by RSS

-p: sort by PSS

-u: sort by USS

-R: convert to [ascending/descending] sort order

-w: Display only the working set statistics counter

-W: Reset the working set statistics counter

-h: Help

For example:

Output memory snapshot:

```
procrank
```

Output memory snapshots in descending order of VSS:

```
procrank –v
```

The default procrank output is sorted by PSS.

11.6.2 Retrieve Specified Content Information

View the memory usage status of a specified process. Command format is as follows:

```
procrank | grep [cmdline | PID]
```

Here, cmdline represents the application name to be searched, and PID represents the process to be searched.

Output the memory usage status of the systemUI process:

```
procrank | grep “com.android.systemui”
```

or:

```
procrank  | grep 3396
```

11.6.3 Tracking Process Memory Status

By tracking memory usage status, it is possible to analyze whether a process exhibits memory leaks. Using scripted approaches to continuously capture process memory snapshots, comparing USS segments reveals if memory leaks are present.

Example: Output the memory usage status of the application with process name com.android.systemui to check for memory leaks:

1. Write script test.sh

```
#!/bin/bash
while true;do
adb shell procrank | grep “com.android.systemui”
sleep 1
done
```

2. After connecting to the device using adb, run this script: ./test.sh

11.7 Dumpsys Tool

The dumpsys tool is a built-in debugging tool in the Android system, running within the device-side shell environment, providing functionality to display status information of services currently running in the system. The running services refer to the server-side processes in the Android Binder mechanism.

Conditions for dumpsys output printing:

1. Only services already loaded into the ServiceManager can be printed.
2. If the dump function in the server-side code is not implemented, no output is generated.

11.7.1 Using Dumpsys

- View dumpsys help

Purpose: Output dumpsys help information.

```
  dumpsys  -help
```

- View the list of services included in Dumpsys

Purpose: Outputs all printable service information from dumpsys, allowing developers to target the names of services to debug.

```
  dumpsys  -l
```

- Output information for the specified service

Purpose: Output dump information for the specified service.

Format: dumpsys [servicename]

Example: Output information about the SurfaceFlinger service. Execute the following command:

```
dumpsys SurfaceFlinger
```

- Output information about the specified service and processes

Function: Output specified service application process information.

Format: dumpsys [servicename] [applicationname]

Example: Output memory information for service meminfo and process com.android.systemui, execute the command:

```
dumpsys meminfo com.android.systemui
```

Note: Service names are case-sensitive and must be entered in full.

11.8 Last log Enable

- Add the following two nodes to the dts file

```
ramoops_mem: ramoops_mem {
    reg = <0x0 0x110000 0x0 0xf0000>;
    reg-names = "ramoops_mem";
};

ramoops {
    compatible = "ramoops";
    record-size = <0x0 0x20000>;
    console-size = <0x0 0x80000>;
    ftrace-size = <0x0 0x00000>;
    pmsg-size = <0x0 0x50000>;
    memory-region = <&ramoops_mem>;
};
```

```
- View last log on the device

130|root@rk3399:/sys/fs/pstore # ls
dmesg-ramoops-0   Saved log after the last kernel panic.
pmsg-ramoops-0    Last user space log, Android log.
ftrace-ramoops-0    prints function tracing within a specific time period.
console-ramoops-0  last_log: the kernel log from the previous boot, but only
retains logs with priority higher than the default log level.
```

- Usage:

```
cat dmesg-ramoops-0
cat console-ramoops-0
logcat -L (pmsg-ramoops-0) pull out by logcat and parse
cat ftrace-ramoops-0
```

11.9 FIQ Mode

When the device freezes or hangs, the system status can be checked by entering fiq command on the serial port. The specific command is as follows:

```
127|console:/ $ fiq
debug>  help
FIQ Debugger commands:
 pc            PC status
 regs          Register dump
 allregs       Extended Register dump
 bt            Stack trace
 reboot [<c>]  Reboot with command <c>
 reset [<c>]   Hard reset with command <c>
 irqs          Interrupt status
 kmsg          Kernel log
 version       Kernel version
 sleep         Allow sleep while in FIQ
 nosleep       Disable sleep while in FIQ
 console       Switch terminal to console
 cpu           Current CPU
 cpu <number>  Switch to CPU<number>
 ps            Process list
 sysrq         sysrq options
 sysrq <param> Execute sysrq with <param>
```

12. Common Issues

12.1 Current kernel and u-boot version?

```
The kernel version corresponding to Android 16.0 is 6.12, and the u-boot branch
is V5.
```

U-Boot Version Verification (via source code):

The U-Boot source code is located in the `u-boot` directory of the SDK.

- **Branch name:** `v5` (manifest specifies `revision="v5"`)
- **U-Boot version:** `2025.04` (based on upstream v2025.04)
- **Recent tag:** `linux-6.12-stan-rkb0-45-gc7d5af87a6a` (with Rockchip platform patches)
- **Remote repository:** `ssh://gerrit-server:29418/android/rk/u-boot`

View local u-boot version info:

```
cd u-boot
git branch -a           # View branch, current should be v5
git describe --tags     # View current tag
grep -E 'VERSION|PATCHLEVEL|SUBLEVEL' Makefile  # View U-Boot version number
```

RK3572 U-Boot defconfig:

```
u-boot/configs/rk3572_defconfig
```

This defconfig uses `rockchip/rk3572-evb` as the default Device Tree, corresponding to `dts/upstream/src/arm64/rockchip/rk3572-evb.dts`.

12.2 How to Obtain the Current SDK's Corresponding RK Release Version

Rockchip Android 16.0 SDK comprises the original AOSP codebase and RK-modified code repositories, with the RK-modified repositories specified in the XML files under the `.repo/manifests/include` directory and the default AOSP repositories defined in `.repo/manifests/default.xml`.

Version Confirmation:

- RK modification section

```
vim .repo/manifests/include/rk_checkout_from_aosp.xml
<project groups="pdk" name="platform/build" path="build/make" remote="rk"
revision="refs/tags/android-16.0-mid-rkr5">
```

The RK version is android-16.0-mid-rkr5

- AOSP Section

```
vim .repo/manifests/default.xml
<default revision="refs/tags/android-16.0.0_r1"...>
```

The AOSP version is android-16.0.0_r1

When version information is required, provide the two version details above.

A single repository can directly retrieve tag information using the following command.

```
kernel-6.12$ git tag
android-16.0-mid-rkr5
```

RK versions increment in the format android-16.0-mid-rkrxx, so the current latest tag is android-16.0-mid-rkr5

12.3 How to Verify the Local SDK Has Been Fully Updated to the RK-Released SDK Status

RK releases SDK versions by committing the corresponding commit information to the .repo/manifests/commit/ directory. Customers can verify whether a full SDK update has been applied by comparing this commit information, as follows:

- Follow the instructions in "How to Obtain the Current SDK's Corresponding RK Release Version" to first confirm the SDK's RK version. The following example uses RK version RKR1 for instruction.
- Use the following command to save local commit information

```
.repo/repo/repo manifest -r -o release_manifest_rkr1_local.xml
```

- By comparing .repo/manifests/commit/commit_release_rkr1.xml and release_manifest_rkr1_local.xml, it can be confirmed whether the SDK code has been fully updated.

The file .repo/manifests/commit/commit_release_rkr1.xml represents the commit information released for RK version RKR1.

12.4 U-Boot and kernel stage logo image replacement

The logo images displayed during the U-Boot and kernel stages respectively correspond to the first and second logo images shown at boot, and can be modified or replaced according to product requirements.

U-Boot logo source file: `kernel-6.12/logo.bmp`

kernel logo source file: `kernel-6.12/logo_kernel.bmp`

To replace a specific image, simply replace it with a BMP file of the same name and recompile the kernel. The compiled file is located in boot.img.

Note: Logo image size is currently limited to 8MB for BMP format images, supporting 8, 16, 24, and 32-bit BMP.

12.5 How to Configure Android to Support Only 64-bit Systems

Starting with Android 14, products passing GMS and EDLA certification must be configured to support only 64-bit systems, with no support for 32-bit architecture. Changing to a 64-bit-only system can reduce memory usage. The specific modifications are as follows (using rk3572_a16 as an example):

Modify the corresponding configuration in the product directory's BoardConfig.mk, as follows:

```
diff --git a/rk3572_a16/BoardConfig.mk b/rk3572_a16/BoardConfig.mk
index c06433e..dc9cc50 100644
--- a/rk3572_a16/BoardConfig.mk
+++ b/rk3572_a16/BoardConfig.mk
@@ -18,3 +18,11 @@ PRODUCT_KERNEL_DTS := rk3572-evb1-v10
 CAMERA_SUPPORT_AUTOFOCUS := true
 BOARD_BUILD_GKI := true
 include device/rockchip/rk3572/rk3572_a16/BoardConfig.mk
+
+DEVICE_IS_64BIT_ONLY := true
+
+TARGET_2ND_ARCH :=
+TARGET_2ND_ARCH_VARIANT :=
+TARGET_2ND_CPU_ABI :=
+TARGET_2ND_CPU_ABI2 :=
+TARGET_2ND_CPU_VARIANT :=
```

12.6 Shutdown Charging and Low-Voltage Precharge

Shutdown charging and low voltage precharge can be configured in the DTS, as follows:

```
charge-animation {
                compatible = "rockchip,uboot-charge";
                rockchip,uboot-charge-on = <1>;
                rockchip,android-charge-on = <0>;
                rockchip,uboot-low-power-voltage = <3400>;
                rockchip,screen-on-voltage = <3500>;
                status = "okay";
        };
```

Specifically:

rockchip,uboot-charge-on: U-Boot shutdown charging is mutually exclusive with Android shutdown charging.

rockchip, android-charge-on: Android shutdown charging is mutually exclusive with U-Boot shutdown charging

rockchip,uboot-low-power-voltage: Configure the precharge voltage from low-voltage state to boot, which can be adjusted based on actual requirements.

rockchip,screen-on-voltage: Configure the precharge voltage from low-power state to screen-on, which can be adjusted based on actual requirements.

12.7 U-Boot Charging Image Packaging and Replacement

Charging image path: the same-named file can be directly replaced, with format requirements matching the original file.

```
u-boot/tools/images/
├── battery_0.bmp
├── battery_1.bmp
├── battery_2.bmp
├── battery_3.bmp
├── battery_4.bmp
├── battery_5.bmp
└── battery_fail.bmp
```

If U-Boot charging is enabled but the charging image does not display, the image may not be packaged in resource.img. Use the following command to package it:

```
cd u-boot
./scripts/pack_resource.sh ../kernel-6.12/resource.img
cp resource.img ../kernel-6.12/resource.img
```

Executing the above command will package the U-Boot charging image into resource.img within the kernel directory. Subsequently, package resource.img into boot.img by executing ./mkimage.sh in the Android root directory, then flash the boot.img located under rockdev.

12.8 HDMI IN Configuration

HDMI In feature in SDK is disabled by default. To enable it, follow the steps below:

```
vim device/rockchip/rk3572/BoardConfig.mk
+BOARD_HDMI_IN_SUPPORT := true
```

12.9 RM310 4G Configuration

4G feature in SDK is disabled by default. To enable it, follow the steps below:

```
vim device/rockchip/common/BoardConfig.mk
#for rk 4g modem
-BOARD_HAS_RK_4G_MODEM ?= false
+BOARD_HAS_RK_4G_MODEM ?= true
```

12.10 Wi-Fi Sleep Policy Configuration

The default Wi-Fi sleep policy maintains connection during sleep. To disconnect during sleep, follow the steps below:

```
---
a/rk3566_rgo/overlay/frameworks/base/packages/SettingsProvider/res/values/defaul
ts.xml
+++
b/rk3566_rgo/overlay/frameworks/base/packages/SettingsProvider/res/values/defaul
ts.xml
@@ -24,5 +24,5 @@
        You can configure persist.wifi.sleep.delay.ms to delay closing wifi.
        The default is 15 minutes, 0 means that the wifi is turned off
        immediately after the screen is off. -->

-    <integer name="def_wifi_sleep_policy">2</integer>

+    <integer name="def_wifi_sleep_policy">0</integer>
 </resources>
```

12.11 Recovery Rotation Configuration

Supports Recovery rotation at 0°/90°/180°/270°, defaulting to 0° (no rotation), with configuration details as follows:

```
vim device/rockchip/common/BoardConfig.mk
#0:   ROTATION_NONE 0-degree rotation
#90: ROTATION_RIGHT rotate 90 degrees
#180: ROTATION_DOWN 180-degree rotation
#270: ROTATION_LEFT 270-degree rotation
# For Recovery Rotation
TARGET_RECOVERY_DEFAULT_ROTATION ?= ROTATION_NONE
```

12.12 Android Surface Rotation

Android system display rotation can be modified by adjusting the following configuration, with configuration parameters set to 0/90/180/270.

```
# For Surface Flinger Rotation
SF_PRIMARY_DISPLAY_ORIENTATION ?= 0
```

12.13 Replace the Remote for Partial AOSP Source Code

Customers experience slow download speeds when fetching RK's release code. To improve speed, modify the AOSP remote to a domestic mirror for customers in China, or to Google's mirror for international customers. The specific procedure is as follows:

After executing repo init (or unpacking the base package), modify .repo/manifests/remote.xml to change the fetch URL for the AOSP remote from

```
<remote name="aosp" fetch="./" review="https://gerrit-server" />
```

to

Domestic customers: (for example, the Tsinghua University mirror source in China, which can be modified to other domestic mirror sources as needed)

```
<remote name="aosp" fetch="https://aosp.tuna.tsinghua.edu.cn" />
```

International customers: (Google mirror source)

```
<remote name="aosp" fetch="https://android.googlesource.com" />
```

12.14 Data Partition Read/Write Rate Optimization

For devices with battery, it is recommended to add the 'fsync_mode=nobarrier' mount option to the data partition in fstab, which can significantly improve storage read/write speed and performance. This parameter poses a risk of data corruption on power loss for devices without battery, thus it is not recommended for devices without battery to add this option. The patch modification is as follows:

```
cd device/rockchip/common

diff --git a/scripts/fstab_tools/fstab.in b/scripts/fstab_tools/fstab.in
index 2ec6c265..c890cc84 100755
--- a/scripts/fstab_tools/fstab.in
+++ b/scripts/fstab_tools/fstab.in
@@ -23,6 +23,6 @@ ${_block_prefix}odm     /odm      ext4 ro,barrier=1
${_flags},first_stage_mount
 # For sdmmc
 /devices/platform/${_sdmmc_device}/mmc_host*        auto  auto    defaults
   voldmanaged=sdcard1:auto
 #  Full disk encryption has less effect on rk3326, so default to enable this.
-/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,keydirectory=/metadata/vold/metadata_encryption,quo
ta,formattable,reservedsize=128M,checkpoint=fs
+/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065,fsync_mode=nobarrier
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,keydirectory=/metadata/vold/metadata_encryption,quo
ta,formattable,reservedsize=128M,checkpoint=fs
 # for ext4
```

```
 #/dev/block/by-name/userdata     /data       ext4
discard,noatime,nosuid,nodev,noauto_da_alloc,data=ordered,user_xattr,barrier=1
latemount,wait,formattable,check,fileencryption=software,quota,reservedsize=128M
,checkpoint=block
diff --git a/scripts/fstab_tools/fstab_go.in b/scripts/fstab_tools/fstab_go.in
index 582557f2..05c7653c 100755
--- a/scripts/fstab_tools/fstab_go.in
+++ b/scripts/fstab_tools/fstab_go.in
@@ -17,6 +17,6 @@ ${_block_prefix}odm     /odm      ext4 ro,barrier=1
${_flags},first_stage_mount
 # For sdmmc
 /devices/platform/${_sdmmc_device}/mmc_host*        auto  auto    defaults
   voldmanaged=sdcard1:auto
 #  Full disk encryption has less effect on rk3326, so default to enable this.
-/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,keydirectory=/metadata/vold/metadata_encryption,quo
ta,formattable,reservedsize=128M,checkpoint=fs
+/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065,fsync_mode=nobarrier
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,keydirectory=/metadata/vold/metadata_encryption,quo
ta,formattable,reservedsize=128M,checkpoint=fs
 # for ext4
 #/dev/block/by-name/userdata     /data       ext4
discard,noatime,nosuid,nodev,noauto_da_alloc,data=ordered,user_xattr,barrier=1
latemount,wait,formattable,check,fileencryption=software,quota,reservedsize=128M
,checkpoint=block
```

12.15 Userdata Partition Filesystem Changed to EXT4

The default file system for the data partition is f2fs. It is recommended that products without battery change the data partition file system to ext4 to minimize the risk of data loss following abnormal power loss. The modification method is as follows:

Taking rk3566_r as an example:

```
device/rockchip/common$ git diff
diff --git a/scripts/fstab_tools/fstab.in b/scripts/fstab_tools/fstab.in
index 6e78b00..a658332 100755
--- a/scripts/fstab_tools/fstab.in
+++ b/scripts/fstab_tools/fstab.in
@@ -20,6 +20,6 @@ ${_block_prefix}system_ext /system_ext  ext4 ro,barrier=1
${_flags},first_stage_
 # For sdmmc
 /devices/platform/${_sdmmc_device}/mmc_host*        auto  auto    defaults
   voldmanaged=sdcard1:auto
 #  Full disk encryption has less effect on rk3326, so default to enable this.
-/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,quota,formattable,reservedsize=128M,checkpoint=fs
```

```
+#/dev/block/by-name/userdata /data f2fs
noatime,nosuid,nodev,discard,reserve_root=32768,resgid=1065
latemount,wait,check,fileencryption=aes-256-xts:aes-256-
cts:v2+inlinecrypt_optimized,quota,formattable,reservedsize=128M,checkpoint=fs
 # for ext4
-#/dev/block/by-name/userdata    /data      ext4
discard,noatime,nosuid,nodev,noauto_da_alloc,data=ordered,user_xattr,barrier=1
latemount,wait,formattable,check,fileencryption=software,quota,reservedsize=128M
,checkpoint=block
+/dev/block/by-name/userdata    /data      ext4
discard,noatime,nosuid,nodev,noauto_da_alloc,data=ordered,user_xattr,barrier=1
latemount,wait,formattable,check,fileencryption=software,quota,reservedsize=128M
,checkpoint=block
```

```
device/rockchip/rk356x$ git diff
diff --git a/rk3566_r/recovery.fstab b/rk3566_r/recovery.fstab
index 7532217..cf789ac 100755
--- a/rk3566_r/recovery.fstab
+++ b/rk3566_r/recovery.fstab
@@ -7,7 +7,7 @@
 /dev/block/by-name/odm                  /odm                ext4
defaults                  defaults
 /dev/block/by-name/cache                /cache              ext4
defaults                  defaults
 /dev/block/by-name/metadata             /metadata           ext4
defaults                  defaults
-/dev/block/by-name/userdata             /data               f2fs
defaults                  defaults
+/dev/block/by-name/userdata             /data               ext4
defaults                  defaults
 /dev/block/by-name/cust                 /cust               ext4
defaults                  defaults
 /dev/block/by-name/custom               /custom             ext4
defaults                  defaults
 /dev/block/by-name/radical_update       /radical_update     ext4
defaults                  defaults
```

12.16 Modify Boot Animation and Boot Sound

Reference Document:

```
RKDocs/android/Rockchip_Introduction_Android_Power_On_Off_Animation_and_Tone_Cus
tomization_CN&EN.pdf
```

12.17 Set Performance Mode for APPs

device/rockchip/rk3xxx/ configuration file: package_performance.xml, add the package names requiring performance mode within the corresponding node: (obtain package name using aapt dump badging (file_path.apk))

```
<app package="package name" mode="1=enabled, 0=disabled"/>
```

For example, the reference for AnTuTu is as follows:

```
<app package="com.antutu.ABenchMark" mode="1"/>
<app package="com.antutu.benchmark.full" mode="1"/>
<app package="com.antutu.benchmark.full" mode="1"/>
```

Files are packaged into the firmware during compilation.

12.18 GPU-Related Troubleshooting Methods

Refer to the following documentation for initial troubleshooting.

```
RKDocs/android/Rockchip_User_Guide_Dr.G_CN&EN.pdf
```

12.19 OTP and efuse Description

OTP-supporting chip

- RK3326
- PX30
- RK3566
- RK3568
- RK3572
- RK3588

EFUSE-enabled chip

- RK3288
- RK3368
- RK3399

Firmware signing and OTP/efuse flashing refer to the document:

```
RKDocs/common/security\Rockchip-Secure-Boot-Application-Note-V1.9.pdf
```

12.20 How to Determine if a Device's OTP/EFUSE Has Been Programmed

OTP/EFUSE status is passed via the kernel cmdline, where the fuse.programmed parameter indicates the OTP/EFUSE status, as follows:

- "fuse.programmed=1": The software firmware package has been signed with secure-boot, and the hardware device's efuse/otp has been programmed.
- "fuse.programmed=0": The software firmware package has been signed for secure-boot, and the hardware device's efuse/otp has not been programmed.
- fuse.programmed is not present in the cmdline: the firmware package was not signed with secure-boot (Miniloader does not propagate), or the Miniloader is too old to support propagation.

12.21 Enabling/Disabling SELinux

The following modification: false = disabled, true = enabled.

```
device/rockchip/common$
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -67,7 +67,7 @@ endif

 # Enable android verified boot 2.0
 BOARD_AVB_ENABLE ?= false
-BOARD_SELINUX_ENFORCING ?= false
+BOARD_SELINUX_ENFORCING ?= true
```

12.22 Boot Warning: "Android system has encountered an issue"

There are two reasons for the appearance of the warning dialog:

1. Firmware mismatch: inconsistent fingerprints for system/boot/vendor, indicating they do not belong to the same set of firmware.
2. When the device is configured with the IO debugging feature enabled, the feature can be disabled by using the kernel build command mentioned earlier.
3. For projects requiring the IO debugging feature, directly apply the patch under frameworks/base to remove the dialog without considering the above two reasons:

```
diff --git
a/services/core/java/com/android/server/wm/ActivityTaskManagerService.java
b/services/core/java/com/android/server/wm/ActivityTaskManagerService.java
index 595c340..d4e495a 100644
--- a/services/core/java/com/android/server/wm/ActivityTaskManagerService.java
+++ b/services/core/java/com/android/server/wm/ActivityTaskManagerService.java
@@ -6555,7 +6555,7 @@ public class ActivityTaskManagerService extends
IActivityTaskManager.Stub {
                 } catch (RemoteException e) {
                 }

-                if (!Build.isBuildConsistent()) {
+                if (0 && !Build.isBuildConsistent()) {
                     Slog.e(TAG, "Build fingerprint is not consistent, warning
user");
                     mUiHandler.post(() -> {
                         if (mShowDialogs) {
```

12.23 How to Access Ethernet Settings in Settings

Ethernet configuration options are not available by default in system settings. If Ethernet is required in the project, it can be enabled as follows:

```
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -146,3 +146,6 @@ endif
 ifeq ($(strip $(BOARD_USES_AB_IMAGE)), true)
 DEVICE_MANIFEST_FILE :=
device/rockchip/$(TARGET_BOARD_PLATFORM)/manifest_ab.xml
 endif

+# for ethernet
+BOARD_HAS_ETHERNET := true
```

12.24 Operations for AVB and Security Boot

Refer to the document for AVB and Security Boot Operations:

```
RKDocs/common/security/Rockchip_Developer_Guide_Android_SecurityBoot_AVB_CN.pdf
```

12.25 Unable to Use IO Commands

IO commands depend on DEVMEM, which is disabled by default, resulting in IO commands being unavailable by default. To use IO commands for debugging, modify as follows:

```
user@ubuntu:~/rk3572/$ vim mkcombinedroot/configs/android-16.config
```

If the product is Go-based, the following modification is required:

```
user@ubuntu:~/rk3572$ vim mkcombinedroot/configs/android-16-go.config
```

Remove the following line:

```
# CONFIG_DEVMEM is not set
```

To build Android, the following modification is also required.

```
cd rk3572/kernel-6.12/kernel/configs

diff --git a/android-6.12/android-base.config b/android-6.12/android-base.config
index 5de76f0..6dcdf86 100644
--- a/android-6.12/android-base.config
+++ b/android-6.12/android-base.config
@@ -2,7 +2,6 @@
 # CONFIG_ANDROID_LOW_MEMORY_KILLER is not set
 # CONFIG_ANDROID_PARANOID_NETWORK is not set
 # CONFIG_BPFILTER is not set
-# CONFIG_DEVMEM is not set
 # CONFIG_FHANDLE is not set
 # CONFIG_FW_CACHE is not set
 # CONFIG_IP6_NF_NAT is not set
user@sys2206:~/rk3572/kernel/configs$ git diff
```

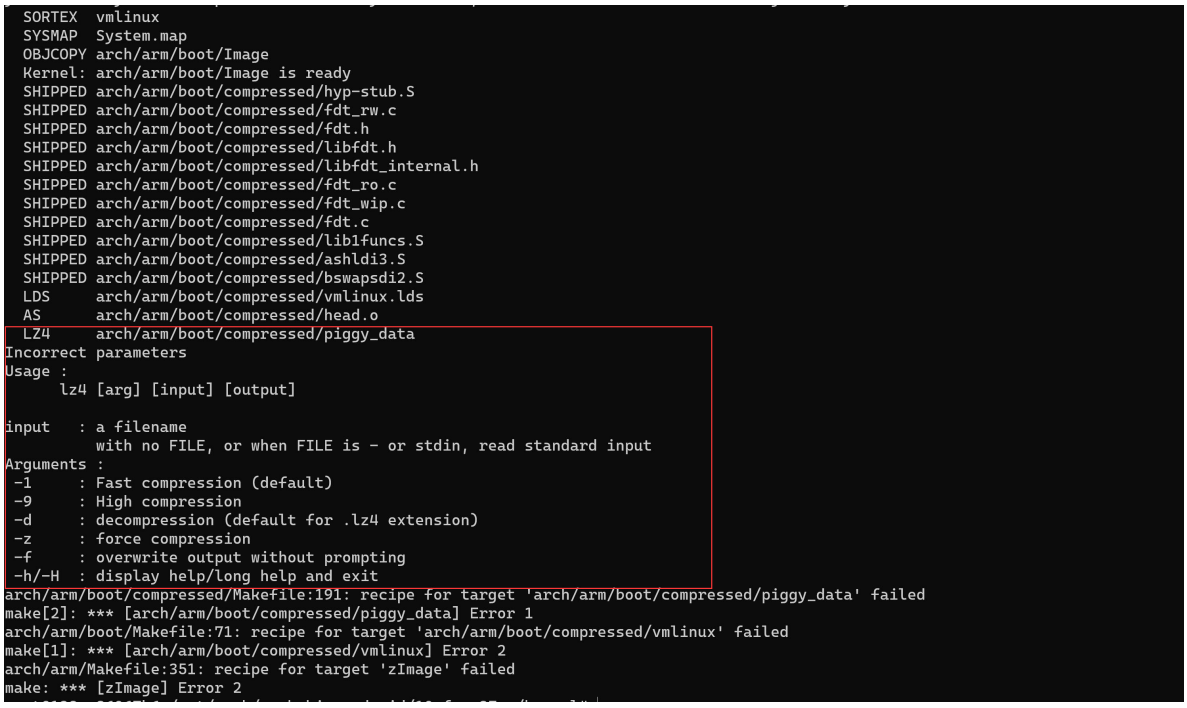
```
diff --git a/u/android-6.12/android-base.config b/u/android-6.12/android-
base.config
index 29b9e98..c1b21cf 100644
--- a/u/android-6.12/android-base.config
+++ b/u/android-6.12/android-base.config
@@ -2,7 +2,6 @@
 # CONFIG_ANDROID_LOW_MEMORY_KILLER is not set
 # CONFIG_ANDROID_PARANOID_NETWORK is not set
 # CONFIG_BPFILTER is not set
-# CONFIG_DEVMEM is not set
 # CONFIG_FHANDLE is not set
 # CONFIG_FW_CACHE is not set
 # CONFIG_IP6_NF_NAT is not set
```

12.26 Naming Rules for SN

SN must start with a letter and not exceed 14 bytes.

12.27 Kernel Compilation Error with LZ4

Kernel compilation reports the following error:



Root Cause:

The system-provided lz4 version is outdated, requiring version 1.8.3 or later.

```
user@ubuntu:~$ lz4 -v
*** LZ4 command line interface 64-bits v1.8.3, by Yann Collet ***
refusing to read from a console
```

Solution:

Directly copy the LZ4 compiled from Android to overwrite the system's LZ4.

```
sudo cp out/host/linux-x86/bin/lz4 /usr/bin/lz4
```

12.28 Android Samba Feature

Reference Document

```
RKDocs/android/Rockchip_Introduction_Android_Samba_CN.pdf
```

12.29 NFS Startup

Reference Documents and Patches:

```
RKDocs/android/patches/customized_functions/nfs_boot_patch_v1.1.0.zip
```

13. Multi-Screen With Different Display and Touch

Reference Document

```
RKDocs/android/patches/customized_functions/Android_Dual_Display_Developer_Guide
.zip
```

14. Multi-Screen Dual Audio

Reference Document

```
  RKDocs/android/patches/customized_functions/Dual_Audio_v1.0.zip
```

15. Appendix A Compiling and Development Environment Setup

15.1 Initializing a Build Environment

This section describes how to set up your local work environment to build the Android source files. You must use Linux or Mac OS; building under Windows is not currently supported.

For an overview of the entire code-review and code-update process, see Life of a Patch.

Note: All commands in this site are preceded by a dollar sign ($) to differentiate them from output or entries within files. You may use the Click to copy feature at the top right of each command box to copy all lines without the dollar signs or triple-click each line to copy it individually without the dollar sign.

15.2 Choosing a Branch

Some requirements for the build environment are determined by the version of the source code you plan to compile. For a full list of available branches, see Build Numbers. You can also choose to download and build the latest source code (called master), in which case you will simply omit the branch specification when you initialize the repository.

After you have selected a branch, follow the appropriate instructions below to set up your build environment.

15.3 Setting up a Linux build environment

These instructions apply to all branches, including master.

The Android build is routinely tested in house on recent versions of Ubuntu LTS (14.04) and Debian testing. Most other distributions should have the required build tools available.

For Gingerbread (2.3.x) and newer versions, including the master branch, a 64-bit environment is required. Older versions can be compiled on 32-bit systems.

Note: See Requirements for the complete list of hardware and software requirements, then follow the detailed instructions for Ubuntu and Mac OS below.

15.4 Installing the JDK

The master branch of Android in the Android Open Source Project (AOSP) comes with prebuilt versions of OpenJDK below prebuilts/jdk/ so no additional installation is required.

Older versions of Android require a separate installation of the JDK. On Ubuntu, use OpenJDK. See JDK Requirements for precise versions and the sections below for instructions.

For Ubuntu >= 15.04

Run the following:

```
sudo apt-get update
sudo apt-get install openjdk-8-jdk
```

For Ubuntu LTS 14.04

There are no available supported OpenJDK 8 packages for Ubuntu 14.04. The Ubuntu 15.04 OpenJDK 8 packages have been used successfully with Ubuntu 14.04. Newer package versions (e.g. those for 15.10, 16.04) were found not to work on 14.04 using the instructions below.

1. Download the .deb packages for 64-bit architecture from old-releases.ubuntu.com:

```
openjdk-8-jre-headless_8u45-b14-1_amd64.deb with SHA256
0f5aba8db39088283b51e00054813063173a4d8809f70033976f83e214ab56c0
openjdk-8-jre_8u45-b14-1_amd64.deb with SHA256
9ef76c4562d39432b69baf6c18f199707c5c56a5b4566847df908b7d74e15849
openjdk-8-jdk_8u45-b14-1_amd64.deb with SHA256
6e47215cf6205aa829e6a0a64985075bd29d1f428a4006a80c9db371c2fc3c4c
```

2. Optionally, confirm the checksums of the downloaded files against the SHA256 string listed with each package above. For example, with the sha256sum tool:

```
    sha256sum {downloaded.deb file}
```

3. Install the packages:

```
    sudo apt-get update
```

Run dpkg for each of the .deb files you downloaded. It may produce errors due to missing dependencies:

```
    sudo dpkg -i {downloaded.deb file}
```

To fix missing dependencies:

```
sudo apt-get -f install
```

Update the default Java version - optional

Optionally, for the Ubuntu versions above update the default Java version by running:

```
sudo update-alternatives --config java
sudo update-alternatives --config javac
```

Note: If, during a build, you encounter version errors for Java, see Wrong Java version for likely causes and solutions.

Installing required packages (Ubuntu 14.04)

You will need a 64-bit version of Ubuntu. Ubuntu 14.04 is recommended.

```
sudo apt-get install git-core gnupg flex bison gperf build-essential zip curl
zlib1g-dev gcc-multilib g++-multilib libc6-dev-i386 lib32ncurses5-dev x11proto-
core-dev libx11-dev lib32z-dev ccache libgl1-mesa-dev libxml2-utils xsltproc
unzip python-pyelftools python3-pyelftools device-tree-compiler libfdt-dev
libfdt1 libssl-dev liblz4-tool python-dev
```

Note: To use SELinux tools for policy analysis, also install the python-networkx package. Note: If you are using LDAP and want to run ART host tests, also install the libnss-sss:i386 package.

15.5 Configuring USB Access

Under GNU/linux systems (and specifically under Ubuntu systems), regular users can't directly access USB devices by default. The system needs to be configured to allow such access.

The recommended approach is to create a file /etc/udev/rules.d/51-android.rules (as the root user) and to copy the following lines in it. must be replaced by the actual username of the user who is authorized to access the phones over USB.

```
# adb protocol on passion (Rockchip products)
SUBSYSTEM=="usb", ATTR{idVendor}=="2207", ATTR{idProduct}=="0010", MODE="0600",
OWNER="<username>"
```

Those new rules take effect the next time a device is plugged in. It might therefore be necessary to unplug the device and plug it back into the computer.

This is known to work on both Ubuntu Hardy Heron (8.04.x LTS) and Lucid Lynx (10.04.x LTS). Other versions of Ubuntu or other variants of GNU/linux might require different configurations.

References : http://source.android.com/source/initializing.html

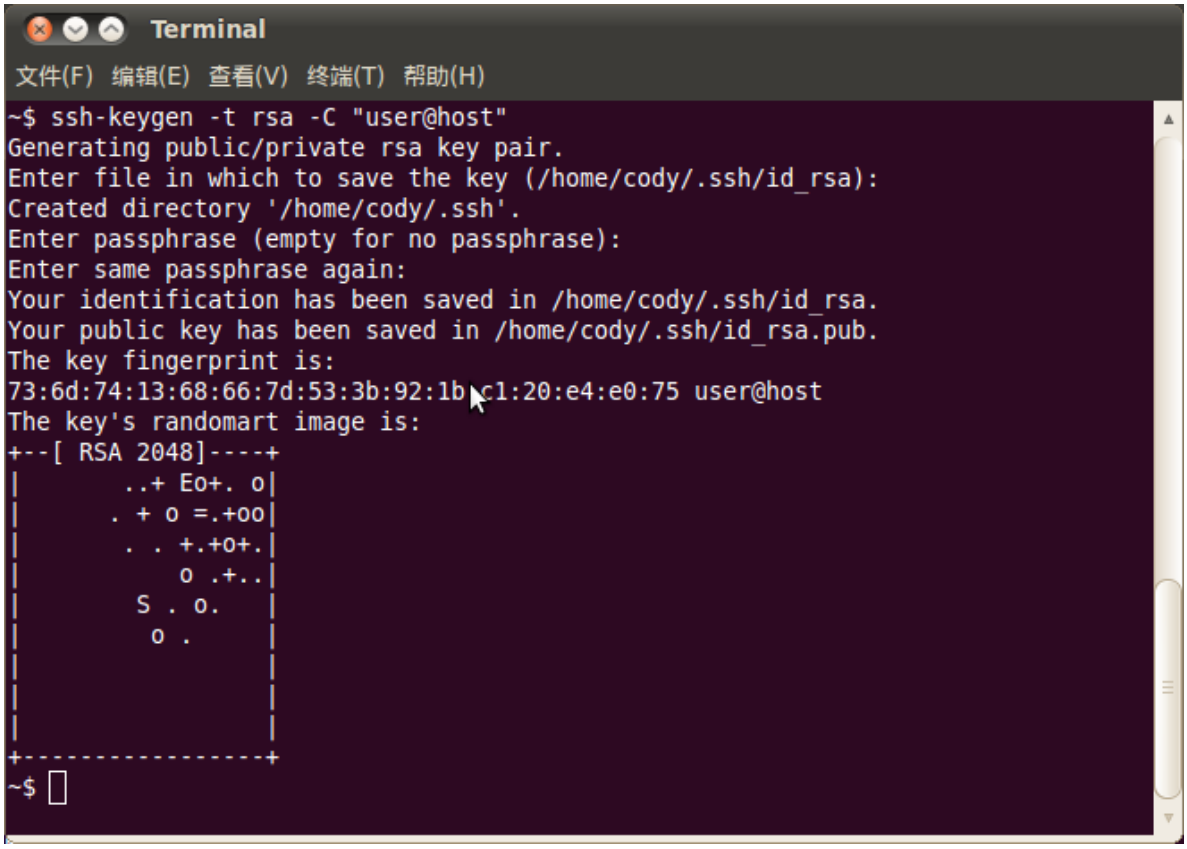
16. Appendix B SSH Public Key Operation Instructions

16.1 Appendix B-1 SSH Public Key Generation

Generate using the following command:

```
ssh-keygen -t rsa -C "user@host"
```

Replace user@host with your email address.



Upon command completion, a key file is generated in the current directory.

```
~$ ls -l .ssh/
总用量 8
-rw------- 1 cody cody 1675 2012-10-15 11:38 id_rsa
-rw-r--r-- 1 cody cody  391 2012-10-15 11:38 id_rsa.pub
```

Please securely store the generated private key file id_rsa and its password, and email the id_rsa.pub file to the administrator of the SDK release server.

16.2 Appendix B-2 Use key-chain to manage the key

We recommend using the simple keychain tool to manage keys.

Detailed usage instructions are as follows:

1. Install the keychain software package:

```
$sudo aptitude install keychain
```

2. Configure to use the key:

```
$vim ~/.bashrc
```

Add the following line:

```
eval `keychain --eval ~/.ssh/id_rsa`
```

id_rsa is the private key file name.

After completing the above configuration, log in to the console again. The system will prompt for a password. Enter the password used during key generation. If no password was set, leave the field blank.

Additionally, avoid using sudo or the root user unless you are aware of how to manage permissions and key management, otherwise it may lead to disorganization.

16.3 Appendix B-3 Multiple devices use the same SSH public key

To use across different devices, copy your SSH private key file id_rsa to `~/.ssh/id_rsa` on the target device.

Using an incorrect private key will display the following prompt. Please replace it with the correct private key.

```
~/tmp$ git clone git@172.16.10.211:rk292x/mid/4.1.1_r1
Initialized empty Git repository in /home/cody/tmp/4.1.1_r1/.git/
The authenticity of host '172.16.10.211 (172.16.10.211)' can't be established.
RSA key fingerprint is fe:36:dd:30:bb:83:73:e1:0b:df:90:e2:73:e4:61:46.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '172.16.10.211' (RSA) to the list of known hosts.
git@172.16.10.211's password:
```

After adding the correct private key, you can clone the code using git, as shown in the figure.

```
~$ cd tmp/
~/tmp$ git clone git@172.16.10.211:rk292x/mid/4.1.1_r1
Initialized empty Git repository in /home/cody/tmp/4.1.1_r1/.git/
The authenticity of host '172.16.10.211 (172.16.10.211)' can't be established.
RSA key fingerprint is fe:36:dd:30:bb:83:73:e1:0b:df:90:e2:73:e4:61:46.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '172.16.10.211' (RSA) to the list of known hosts.
remote: Counting objects: 237923, done.
remote: Compressing objects: 100% (168382/168382), done.
Receiving objects:   9% (21570/237923), 61.52 MiB | 11.14 MiB/s
```

Adding an SSH private key may display the following error message.

```
Agent admitted failure to sign using the key
```

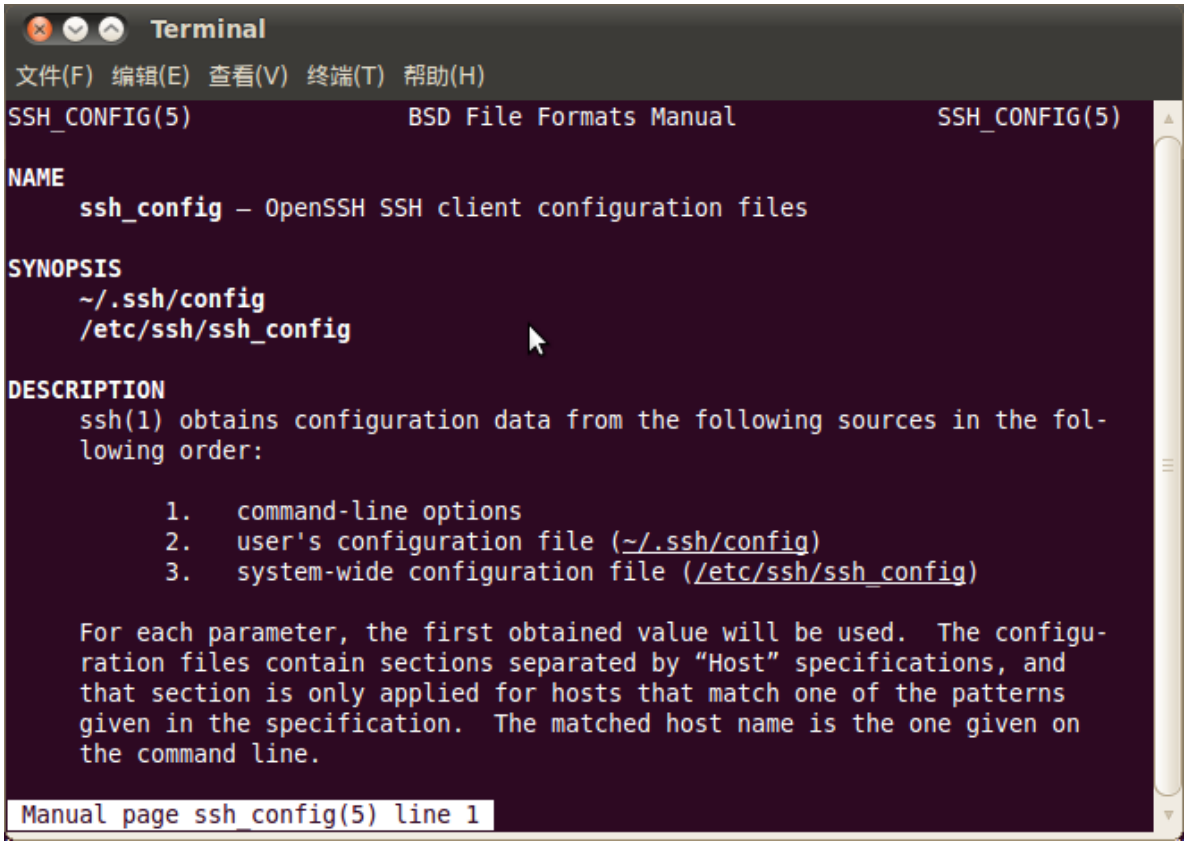
Enter the following command in the console to resolve the issue.

```
ssh-add ~/.ssh/id_rsa
```

16.4 Appendix B-4 Switching Different SSH Public Keys on a Single Device

Refer to the ssh_config file for SSH configuration.

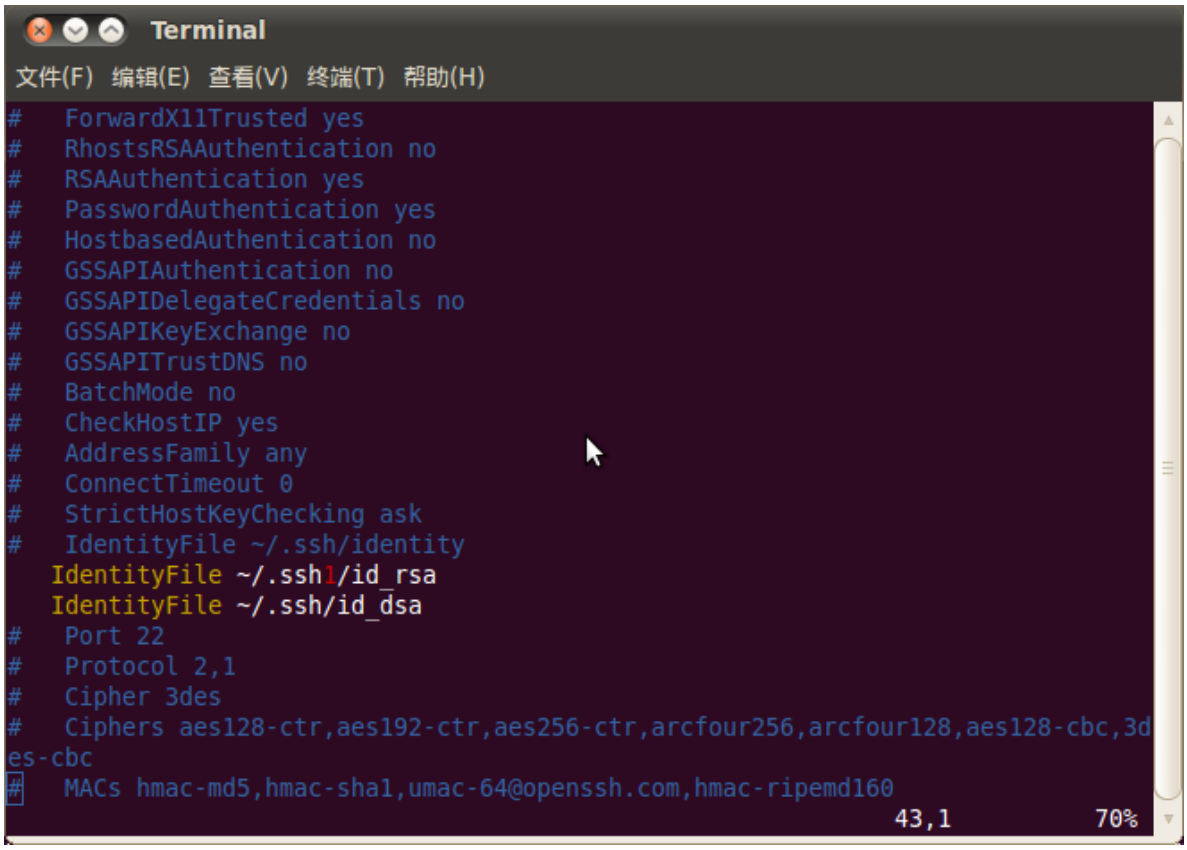
```
~$ man ssh_config
```



Use the following command to configure SSH for the current user.

```
~$ cp /etc/ssh/ssh_config ~/.ssh/config
~$ vi .ssh/config
```

As shown in the figure, configure SSH to use the file `~/.ssh1/id_rsa` as the private key. This method allows switching between different keys.



16.5 Appendix B-5 Key Authority Management

The server can monitor the download count, IP addresses, and other information in real time for a specific key. If anomalies are detected, the download permissions for the corresponding key will be disabled.

Store the private key file securely and do not re-authorize for third-party use.

16.6 Appendix B-6 Git Authority Application Instruction

Refer to the preceding section, generate a public key file, and send an email to fae@rock-chips.com to request access to SDK code downloads.